**Advanced Thermodynamics**

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**Final Project**

# Analysis of organic Rankine cycle with internal heat exchanger


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# Abstract

This project presents a complete thermodynamic, exergetic, heat transfer, economic, and parametric analysis of a geothermal Organic Rankine Cycle with an Internal Heat Exchanger. The cycle was simulated in EES using R123 as the working fluid and a geothermal source temperature of $180°C$. The base operating conditions included a condenser temperature of $40°C$, an evaporator temperature of $120°C$, a pinch point temperature difference of $10°C$, a turbine isentropic efficiency of 0.8, a pump isentropic efficiency of 0.9, and an internal heat exchanger effectiveness of 0.6.

The thermodynamic model was first developed to determine the temperature, pressure, specific enthalpy, and specific entropy at all state points. The base-case results showed a net power output of 49.54 kW, a working fluid mass flow rate of 1.799 kg/s, a thermal efficiency of 13.78%, and an exergy efficiency of 38.17%. The total exergy destruction rate was calculated as 80.25 kW. The largest exergy loss was associated with the reinjected geothermal fluid, followed by the evaporator, turbine, and condenser.

The heat exchanger area calculation was performed using the LMTD method. The required areas of the condenser, evaporator, and internal heat exchanger were obtained as 8.04 $m^2$, and 5.63 $m^2$, respectively. A thermo-economic analysis was also carried out using the cost correlations from Paper 2[1]. The total product cost rate was calculated as 0.001651 \$/s, equivalent to 5.944 \$/h, and the specific production cost was 33.33 \$/GJ.

Parametric studies were conducted by varying the pinch point temperature difference, evaporator temperature, and condenser temperature. The results showed clear trade-offs between thermodynamic performance, heat exchanger size, and economic cost. Finally, the simulation results were validated against Paper 1[2]. The close agreement between the present results and the reference data confirmed the reliability of the developed EES model

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# Introduction

Organic Rankine Cycles are widely used for converting low- and medium-temperature heat sources into useful power. In this project, a geothermal Organic Rankine Cycle with an Internal Heat Exchanger was modeled and analyzed. The cycle uses R123 as the working fluid and receives heat from a geothermal fluid stream. The main purpose of the project was to simulate the ORC-IHE system, evaluate its thermodynamic and exergetic performance, estimate the required heat exchanger areas, perform a thermo-economic analysis, investigate the effect of important design parameters, and validate the results against the reference data reported in Paper 1[2].

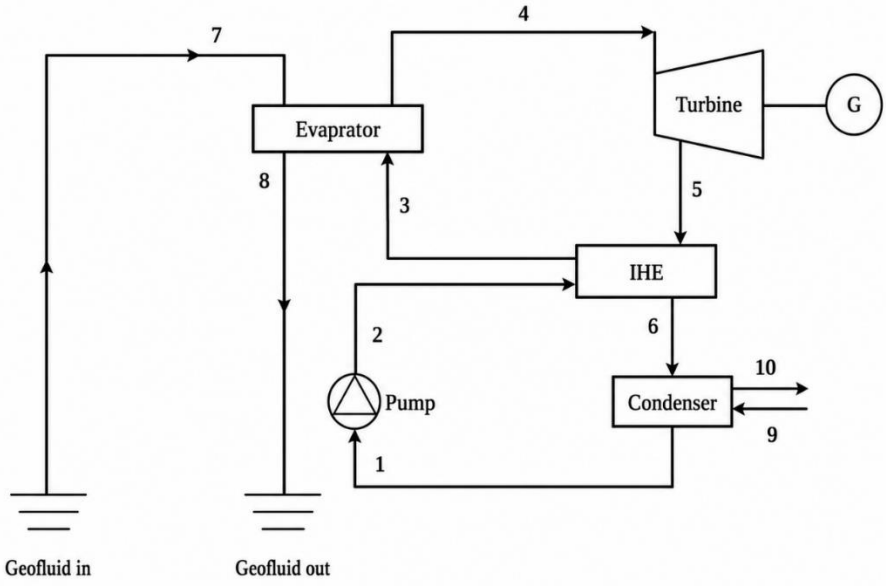



*Figure 1: Schematic diagram of the Organic Rankine Cycle with Internal Heat Exchanger*

According to the final project instructions, the ORC with IHE must be simulated in EES using the input data given in Table 1 and the modeling equations provided in Paper 1[2]. The required input parameters include the dead-state conditions, condenser temperature, evaporator temperature, geothermal inlet temperature, pinch point temperature difference, turbine efficiency, pump efficiency, working fluid, and IHE effectiveness. The base case of the project is defined by $P_0 = 100\ \text{kPa}, T_0 = 25°C, T_{cond} = 40°C, T_{evp} = 120°C, T_{geo} = 180°C, \Delta T_{pinch} = 10°C, \eta_T = 0.8, \eta_P = 0.9$, R123 as the working fluid, and $\varepsilon_{IHE} = 0.6$.

The system consists of a pump, evaporator, turbine, internal heat exchanger, condenser, geothermal fluid loop, and cooling air stream. In the ORC loop, the working fluid leaves the condenser as saturated liquid and is compressed by the pump to the evaporator pressure. Before

entering the evaporator, the working fluid is preheated in the internal heat exchanger using the turbine exhaust stream. The working fluid then receives heat from the geothermal fluid in the evaporator and enters the turbine as saturated vapor. After expansion in the turbine, the exhaust vapor transfers part of its remaining thermal energy to the pressurized liquid in the IHE before entering the condenser. This internal heat recovery improves the thermal utilization of the working fluid and reduces the heat rejection load in the condenser.

The first part of the project focused on thermodynamic modeling and energy-exergy analysis. The ORC-IHE cycle was implemented in EES, and the thermodynamic properties at all state points were calculated. These properties included temperature, pressure, specific enthalpy, and specific entropy. The EES model was then used to calculate the main performance parameters of the system, including thermal efficiency, exergy efficiency, working fluid mass flow rate, total exergy destruction rate, and net power output. In addition, the exergy destruction rate of each component was calculated and plotted as a bar chart to identify the major sources of irreversibility in the system.

The second part of the project involved heat exchanger area calculation. The required areas of the condenser, evaporator, and internal heat exchanger were calculated using the logarithmic mean temperature difference method. The project statement provides the heat transfer relation:

$$\dot{Q} = UA\Delta T_{LMTD} \tag{1}$$

and the LMTD relation:

$$\Delta T_{LMTD} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)} \tag{2}$$

The overall heat transfer coefficients were taken from Table 2 of the project statement. The values used were $250\ \mathrm{W/m^2.K}$ for the condenser, $300\ \mathrm{W/m^2.K}$ for the internal heat exchanger, and $850\ \mathrm{W/m^2.K}$ for the evaporator.

The third part of the project was the economic analysis. Using the equations and cost data provided in Paper 2[1], a thermo-economic analysis was performed for the simulated ORC-IHE cycle. The total product cost rate was calculated by considering the fuel cost rate associated with the geothermal exergy input and the annualized capital cost rates of the main components. The capital investment costs of the turbine, pump, condenser, evaporator, and internal heat exchanger

were calculated using the cost correlations from Paper 2[1]. The final economic results included the total product cost rate, expressed in \$/s and \$/h, and the specific production cost, expressed in \$/GJ and \$/kWh.

The fourth part of the project consisted of parametric studies. Three important design parameters were varied to evaluate their effects on the ORC-IHE performance. First, the pinch point temperature difference was varied from $5°C$ to $20°C$, and its effects on evaporator area, working fluid mass flow rate, and exergy efficiency were studied. Second, the evaporator temperature was varied from $120°C$ to $140°C$, and its effects on thermal efficiency and exergy efficiency were analyzed. Third, the condenser temperature was varied from $30°C$ to $50°C$, and its effects on thermal efficiency, exergy efficiency, and total product cost rate were investigated. These parametric studies were used to show the trade-offs between thermodynamic performance, heat exchanger sizing, and economic behavior of the system.

The final part of the project was validation. The results obtained from the EES model were compared with the data and findings reported in Paper 1[2], as required in the project statement. The validation included comparison of state properties, net power output, thermal efficiency, exergy efficiency, total exergy destruction rate, and the distribution of exergy destruction among the components. The comparison showed that the present model reproduces the reference results with good accuracy. The total exergy destruction rate matched the value reported in Paper 1[2], and the main performance parameters showed only small deviations. Therefore, the developed EES model was considered reliable for the heat exchanger area calculation, economic analysis, and parametric studies.

Overall, this project provides a complete thermodynamic, exergetic, heat transfer, economic, and parametric assessment of a geothermal ORC with an Internal Heat Exchanger. The study identifies the main sources of irreversibility, evaluates the required heat exchanger sizes, estimates the cost of power production, and examines how key operating parameters affect the performance of the cycle. The results show that the ORC-IHE system performance is strongly influenced by the pinch point temperature difference, evaporator temperature, and condenser temperature, and that the final design must consider both thermodynamic efficiency and economic cost.

# Part A: Thermodynamic Modeling (Energy + Exergy Analysis)

In this part, the Organic Rankine Cycle (ORC) equipped with an Internal Heat Exchanger (IHE) is simulated using Engineering Equation Solver (EES). The main objective is to model the thermodynamic behavior of the cycle based on the input data given in Table 1 and the governing equations presented in Paper 1[2].

The cycle uses R123 as the working fluid. The geothermal heat source provides thermal energy to the evaporator, where the working fluid is heated and vaporized. The vapor then expands through the turbine to produce power. After leaving the turbine, the working fluid passes through the internal heat exchanger, where part of its remaining thermal energy is recovered to preheat the compressed liquid before entering the evaporator. Finally, the working fluid is condensed in the condenser and pumped back to the high-pressure side of the cycle.

Before solving the subparts A_1 to A_4, all required input parameters are defined. These parameters include the ambient conditions, condenser and evaporator temperatures, geothermal source temperature, pinch-point temperature difference, turbine and pump efficiencies, working fluid, and internal heat exchanger effectiveness.

*Table 1: The input data used for the EES simulation.*

| Parameter | Value | Unit |
|---|---|---|
| $(P_0)$ | 100 | kPa |
| $(T_0)$ | 25 | °C |
| $(T_{cond})$ | 40 | °C |
| $(T_{evp})$ | 120 | °C |
| $(T_{geo})$ | 180 | °C |
| $(\Delta T_{pinch})$ | 10 | °C |
| $(\eta_T)$ | 0.8 | – |
| $(\eta_P)$ | 0.9 | – |
| **Working Fluid** | R123 | – |
| $(\varepsilon_{IHE})$ | 0.6 | – |

## 1-1 A_1:Evaluation of Thermodynamic Properties at All ORC State Points

### 1-1-1 Objective

The objective of this section is to simulate the Organic Rankine Cycle with an Internal Heat Exchanger using EES and to determine the thermodynamic properties at all state points of the cycle. The required properties include temperature, pressure, specific enthalpy, and specific entropy. The system is modeled based on the ORC with IHE configuration shown in Paper 1[2]. In this configuration, the internal heat exchanger is used to recover part of the thermal energy of the turbine exhaust and preheat the working fluid before it enters the evaporator.

The working fluid selected for the ORC is R123. The geothermal fluid is modeled using water properties, as assumed in Paper 1[2]. The condenser temperature, evaporator temperature, geothermal inlet temperature, pinch-point temperature difference, pump efficiency, turbine efficiency, and IHE effectiveness are taken from the given input table.

### 1-1-2 Input Data

*Table 2:Input data for EES code of section A_1*

| Parameter | Value | Unit |
|---|---|---|
| $P_0$ | 100 | kPa |
| $T_0$ | 25 | °C |
| $T_{cond}$ | 40 | °C |
| $T_{rvp}$ | 120 | °C |
| $T_{geo}$ | 180 | °C |
| $\Delta T_{pinch}$ | 10 | °C |
| $\eta_T$ | 0.80 | – |
| $\eta_P$ | 0.90 | – |
| **Working Fluid** | R123 | – |
| $\varepsilon_{IHE}$ | 0.60 | – |

### 1-1-3 Simplifying Assumptions

The following assumptions were used to simplify the thermodynamic model:

1. The cycle operates under steady-state conditions.
2. Pressure drops in the evaporator, condenser, internal heat exchanger, and connecting pipes are neglected.
3. The pump and turbine are modeled using isentropic efficiencies.
4. Kinetic and potential energy changes are neglected.
5. The geofluid is assumed to be saturated liquid at the geothermal inlet condition.
6. Water properties are used for the geofluid.
7. The working fluid leaves the condenser as saturated liquid.
8. The working fluid leaves the evaporator as saturated vapor.

9. The internal heat exchanger is considered adiabatic to the surroundings.
10. Heat losses from the evaporator, condenser, turbine, pump, and IHE to the environment are neglected.

The first six assumptions are consistent with the modeling assumptions reported in Paper 1[2], where the authors considered steady-state operation, neglected pressure losses, used isentropic efficiencies for turbomachinery, neglected kinetic and potential energy changes, assumed saturated-liquid geofluid in the reservoir, and used water properties for the geofluid.

*Table 3: State Point Description*

| State | Description |
|---|---|
| **1** | Condenser outlet / Pump inlet |
| **2** | Pump outlet / IHE cold – side inlet |
| **3** | IHE cold – side outlet / Evaporator inlet |
| **4** | Evaporator outlet / Turbine inlet |
| **5** | Turbine outlet / IHE hot – side inlet |
| **6** | IHE hot – side outlet / Condenser inlet |
| **7** | Geofluid inlet |
| **8** | Geofluid outlet |
| **9** | Cooling air inlet |
| **10** | Cooling air outlet |

## 1-1-4 Thermodynamic Modeling Equations

### 1-1-4-1 Condenser Pressure and Evaporator Pressure

The condenser pressure is obtained from the saturation pressure of R123 at the condenser temperature:

$$P_{cond} = P_{sat,R123}(T_{cond}) \tag{3}$$

$$P_1 = P_5 = P_6 = P_{cond} \tag{4}$$

From the EES output:

$$P_{cond} = 154.7\ \text{kPa} \tag{5}$$

The evaporator pressure is obtained from the saturation pressure of R123 at the evaporator temperature:

$$P_{evp} = P_{sat,R123}(T_{evp}) \tag{6}$$

$$P_2 = P_3 = P_4 = P_{evp}$$

$$P_{evp} = 1201\ \text{kPa} \tag{7}$$

From the EES output:

### 1-1-4-2 Pump Model

At state 1, the working fluid leaves the condenser as saturated liquid. The pump increases the pressure from condenser pressure to evaporator pressure.

The ideal pump outlet enthalpy is calculated as:

$$h_{2s} = h_1 + v_1(P_2 - P_1) \tag{8}$$

The pump isentropic efficiency is defined as:

$$h_{2s} = h_1 + v_1(P_2 - P_1) \tag{9}$$

$$\eta_P = \frac{h_{2s} - h_1}{h_2 - h_1} \tag{10}$$

Therefore, the actual pump outlet enthalpy is:

$$h_2 = h_1 + \frac{h_{2s} - h_1}{\eta_P} \tag{11}$$

The pump power consumption is:

$$\dot{W}_P = \dot{m_{wf}}(h_2 - h_1) \tag{12}$$

Using the EES results:

$$\dot{W}_P = 1.468\ \text{kW} \tag{13}$$

The small pump work is expected because the working fluid is in liquid phase during compression.

### 1-1-4-3 Internal Heat Exchanger Model

The internal heat exchanger transfers heat from the hot turbine exhaust stream to the pressurized liquid stream leaving the pump. In this model, states 5 and 6 represent the hot side of the IHE, while states 2 and 3 represent the cold side.

The IHE effectiveness is written as:

$$\varepsilon_{IHE} = \frac{T_5 - T_6}{T_5 - T_2} \tag{14}$$

Thus, the hot-side outlet temperature is:

$$T_6 = T_5 - \varepsilon_{IHE}(T_5 - T_2) \tag{15}$$

The energy balance across the IHE is:

$$\dot{Q_{IHE}} = \dot{m_{wf}}(h_5 - h_6) \tag{16}$$

Since the same working-fluid mass flow rate passes through both sides of the IHE:

$$h_3 = h_2 + h_5 - h_6 \tag{17}$$

From the EES output:

$$\dot{Q_{IHE}} = 18.54\ \text{kW} \tag{18}$$

The IHE increases the temperature of the working fluid from $40.48°C$ at state 2 to $50.19°C$ at state 3. This reduces the required external heat addition in the evaporator.

### 1-1-4-4 Evaporator Model

In the evaporator, heat is transferred from the geothermal fluid to the ORC working fluid. The working fluid enters the evaporator at state 3 and leaves as saturated vapor at state 4.

The evaporator heat transfer rate is:

$$\dot{Q_{evap}} = \dot{m_{wf}}(h_4 - h_3) \tag{19}$$

The same heat transfer is equal to the heat lost by the geofluid:

$$\dot{Q_{evap}} = \dot{m_{geo}}(h_7 - h_8) \quad (20)$$

The pinch-point condition is defined as:

$$\Delta T_{pinch} = T_{pp} - T_{evp} \quad (21)$$

Therefore:

$$T_{pp} = T_{evp} + \Delta T_{pinch} \quad (22)$$

Using the input values:

$$T_{pp} = 120 + 10 = 130°C \quad (23)$$

In the EES model, the working-fluid mass flow rate is determined using the pinch-point energy balance:

$$\dot{m_{wf}}(h_4 - h_{f,evp}) = \dot{m_{geo}}(h_7 - h_{pp}) \quad (24)$$

where $hf,evph_{f,evp}hf,evp$ is the saturated-liquid enthalpy of R123 at the evaporator temperature, and hpph_{pp}hpp is the geofluid enthalpy at the pinch-point temperature.

From the EES output:

$$\dot{m_{wf}} = 1.799 \text{ kg/s} \quad (25)$$

$$\dot{Q_{evap}} = 359.6 \text{ kW} \quad (26)$$

The geofluid enters the evaporator at $180°C$ and leaves at $96.11°C$. This temperature drop shows that a significant amount of geothermal energy is transferred to the ORC working fluid.

### 1-1-4-5 Turbine Model

The working fluid enters the turbine at state 4 as saturated vapor at the evaporator pressure and expands to the condenser pressure at state 5.

For the isentropic turbine expansion:

$$s_{5s} = s_4 \quad (27)$$

The ideal turbine outlet enthalpy is:

$$h_{5s} = h(P_5, s_{5s}) \quad (28)$$

The turbine isentropic efficiency is:

$$\eta_T = \frac{h_4 - h_5}{h_4 - h_{5s}} \quad (29)$$

Therefore, the actual turbine outlet enthalpy is:

$$h_5 = h_4 - \eta_T(h_4 - h_{5s}) \quad (30)$$

The turbine power output is:

$$\dot{W_T} = \dot{m_{wf}}(h_4 - h_5) \quad (31)$$

From the EES output:

$$\dot{W_T} = 51.01 \text{ kW} \quad (32)$$

The turbine outlet temperature is $63.27°C$, which is still higher than the pump outlet temperature. Therefore, heat recovery in the IHE is thermodynamically possible.

### 1-1-4-6 Condenser Model

After leaving the IHE, the working fluid enters the condenser at state 6 and leaves as saturated liquid at state 1.

The condenser heat rejection rate is:

$$\dot{Q_{cond}} = \dot{m_{wf}}(h_6 - h_1) \tag{33}$$

The condenser heat rejection is absorbed by the cooling air:

$$\dot{Q_{cond}} = \dot{m_{air}}(h_{10} - h_9) \tag{34}$$

Therefore, the required cooling air mass flow rate is:

$$\dot{m_{air}} = \frac{\dot{Q_{cond}}}{h_{10} - h_9} \tag{35}$$

From the EES output:

$$\dot{Q_{cond}} = 310.1 \text{ kW} \tag{36}$$

$$\dot{m}air = 30.85\ kg/s \tag{37}$$

The cooling air enters the condenser at $25°C$ and exits at $35°C$.

### 1-1-4-7 Net Power Output and ORC Efficiency

The net power output of the ORC is calculated as:

$$\dot{W_{net}} = \dot{W}_T - \dot{W}_P \tag{38}$$

Using the EES results:

$$\dot{W_{net}} = 51.01 - 1.468 = 49.54 \text{ kW} \tag{39}$$

The ORC thermal efficiency based on the heat input to the ORC is:

$$\eta_{ORC} = \frac{\dot{W_{net}}}{\dot{Q_{ORC}}} \tag{40}$$

where:

$$\dot{Q_{ORC}} = \dot{Q_{evap}} \tag{41}$$

Therefore:

$$\eta_{ORC} = \frac{49.54}{359.6} = 0.1378 = 13.78\% \tag{42}$$

This efficiency is based only on the heat transferred to the ORC working fluid in the evaporator. It should not be confused with the overall geothermal plant efficiency based on the total geothermal energy input.

## 1-1-5 EES Numerical Results

*Table 4:Main Cycle Results of section A_1*

| Parameter | Value | Unit |
|---|---|---|
| $\dot{m}_{geo}$ | 1.000 | $kg/s$ |
| $\dot{m}_{wf}$ | 1.799 | $kg/s$ |
| $\dot{m}_{air}$ | 30.85 | $kg/s$ |
| $P_{cond}$ | 154.7 | $kPa$ |
| $P_{evp}$ | 1201 | $kPa$ |
| $\dot{Q}_{evap}$ | 359.6 | $kW$ |
| $\dot{Q}_{IHE}$ | 18.54 | $kW$ |
| $\dot{Q}_{cond}$ | 310.1 | $kW$ |
| $\dot{W}_T$ | 51.01 | $kW$ |
| $\dot{W}_P$ | 1.468 | $kW$ |
| $\dot{W}_{net}$ | 49.54 | $kW$ |
| $\eta_{ORC}$ | 13.78 | % |

*Table 5: Thermodynamic Properties at State Points of section A_1*

| State | Fluid | $T\ (°C)$ | $P\ (kPa)$ | $h\ (kJ/kg)$ | $s\ (kJ/kg \cdot K)$ |
|---|---|---|---|---|---|
| **1** | $R123$ | 40.00 | 154.7 | 241.9 | 1.142 |
| **2** | $R123$ | 40.48 | 1201 | 242.7 | 1.143 |
| **3** | $R123$ | 50.19 | 1201 | 253.0 | 1.175 |
| **4** | $R123$ | 120.00 | 1201 | 452.9 | 1.702 |
| **5** | $R123$ | 63.27 | 154.7 | 424.6 | 1.724 |
| **6** | $R123$ | 49.60 | 154.7 | 414.3 | 1.693 |
| **7** | $Geofluid$ | 180.00 | 1003 | 763.1 | 2.139 |
| **8** | $Geofluid$ | 96.11 | 1003 | 403.4 | 1.262 |
| **9** | $Air$ | 25.00 | 100 | 298.6 | 5.699 |
| **10** | $Air$ | 35.00 | 100 | 308.6 | 5.733 |

## 1-1-6 Discussion of Results

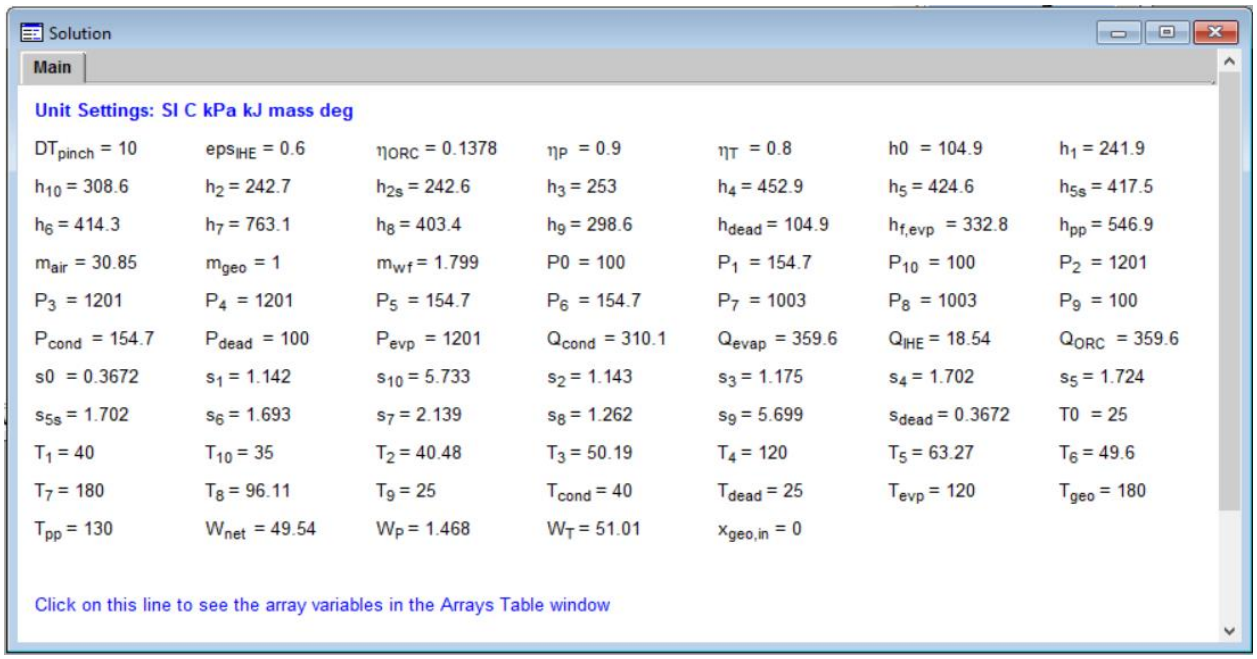



*Figure 2: EES results for part A_1*

The EES simulation shows that the ORC with IHE operates between a condenser pressure of 154.7kPa and an evaporator pressure of 1201kPa. The working fluid leaves the condenser as saturated liquid at $40°C$, is compressed by the pump, and then enters the IHE. Since the pump work is only 1.468 kW, it is small compared with the turbine output.

The IHE increases the working-fluid temperature from $40.48°C$ to $50.19°C$ before the evaporator. This preheating effect reduces the external heat demand of the evaporator. The heat recovered in the IHE is 18.54 kW. Although this value is smaller than the evaporator heat input, it improves heat utilization within the cycle.

The evaporator supplies 359.6 kW of heat to the working fluid. The working fluid exits the evaporator at $120°C$ and 1201 kPa as saturated vapor. It then expands through the turbine and produces 51.01kW of power. After expansion, the turbine outlet still has a relatively high temperature of $63.27°C$, which makes internal heat recovery possible.

The condenser rejects 310.1310.1310.1 kW of heat to the cooling air. For a cooling air temperature rise from $25°C$ to $35°C$, the required air mass flow rate is 30.85 kg/s. The relatively large cooling air flow rate is due to the small temperature rise allowed for the cooling air.

The net power output of the system is 49.54kW. The thermal efficiency of the ORC based on the evaporator heat input is 13.78%. This value represents the efficiency of the organic Rankine cycle itself and not the overall geothermal power plant efficiency based on the total energy content of the geothermal fluid.

*Table 6: Thermodynamic Properties at All State Points based on EES results for paer A_1*

| State | Fluid | $T\ (°C)$ | $P\ (kPa)$ | $h\ (kJ/kg)$ | $s\ (kJ/kg \cdot K)$ |
|---|---|---|---|---|---|
| **1** | $R123$ | 40.00 | 154.7 | 241.9 | 1.142 |

| | | | | | |
|---|---|---|---|---|---|
| **2** | *R123* | 40.48 | 1201 | 242.7 | 1.143 |
| **3** | *R123* | 50.19 | 1201 | 253.0 | 1.175 |
| **4** | *R123* | 120.00 | 1201 | 452.9 | 1.702 |
| **5** | *R123* | 63.27 | 154.7 | 424.6 | 1.724 |
| **6** | *R123* | 49.60 | 154.7 | 414.3 | 1.693 |
| **7** | *Geofluid* | 180.00 | 1003 | 763.1 | 2.139 |
| **8** | *Geofluid* | 96.11 | 1003 | 403.4 | 1.262 |
| **9** | *Air* | 25.00 | 100 | 298.6 | 5.699 |
| **10** | *Air* | 35.00 | 100 | 308.6 | 5.733 |

*Table 7: Additional EES Results for part A_1*

| Parameter | Value | Unit |
|---|---|---|
| $(\boldsymbol{m_{geo}})$ | 1.000 | $kg/s$ |
| $(\boldsymbol{m_{wf}})$ | 1.799 | $kg/s$ |
| $(\boldsymbol{m_{air}})$ | 30.85 | $kg/s$ |
| $(\boldsymbol{P_{cond}})$ | 154.7 | $kPa$ |
| $(\boldsymbol{P_{evap}})$ | 1201 | $kPa$ |
| $(\boldsymbol{Q_{evap}})$ | 359.6 | $kW$ |
| $(\boldsymbol{Q_{IHE}})$ | 18.54 | $kW$ |
| $(\boldsymbol{Q_{cond}})$ | 310.1 | $kW$ |
| $(\boldsymbol{W_T})$ | 51.01 | $kW$ |
| $(\boldsymbol{W_P})$ | 1.468 | $kW$ |
| $(\boldsymbol{W_{net}})$ | 49.54 | $kW$ |
| $(\boldsymbol{\eta_{ORC}})$ | 13.78 | % |

## 1-2 A_2: Performance Evaluation of the ORC with Internal Heat Exchanger

### 1-2-1 Objective

The objective of this section is to evaluate the overall performance of the Organic Rankine Cycle with an Internal Heat Exchanger using the thermodynamic properties obtained from the EES simulation in Part A-1. The required performance parameters are thermal efficiency, exergy efficiency, working fluid mass flow rate, total exergy destruction rate, and net power output.

The cycle is modeled as a binary geothermal power plant using R123 as the working fluid. The geothermal fluid transfers heat to the ORC working fluid in the evaporator, while the internal heat exchanger recovers part of the thermal energy from the turbine exhaust before condensation. The energy and exergy modeling approach follows the relations used in Paper 1[2], where the cycle is evaluated based on first-law efficiency, second-law efficiency, specific flow exergy, and exergy destruction rate.

### 1-2-2 Performance Summary Table

*Table 8: Input data for EES code of section A_2*

| Performance Parameter | Symbol | Value | Unit |
|---|---|---|---|
| **Thermal efficiency** | $\eta_{th}$ | 13.78 | % |
| **Exergy efficiency** | $\eta_{ex}$ | 38.17 | % |
| **Working fluid mass flow rate** | $\dot{m}_{wf}$ | 1.799 | kg/s |
| **Total exergy destruction rate** | $\dot{E}_{D,total}$ | 80.25 | kW |
| **Net power output** | $\dot{W}_{net}$ | 49.54 | kW |

### 1-2-3 Governing Equations

#### 1-2-3-1 Turbine Power Output

The turbine power output is calculated from the enthalpy drop of the working fluid during expansion:

$$\dot{W}_T = \dot{m_{wf}}(h_4 - h_5) \tag{43}$$

From the EES results:

$$\dot{W}_T = 51.01\ \text{kW} \tag{44}$$

The turbine is the main power-producing component of the ORC. The working fluid enters the turbine at state 4 and exits at state 5 after expansion to the condenser pressure.

### 1-2-3-2 Pump Power Consumption

The pump power input is calculated from the enthalpy rise of the working fluid across the pump:

$$\dot{W}_P = \dot{m_{wf}}(h_2 - h_1) \tag{45}$$

From the EES results:

$$\dot{W}_P = 1.468\ \text{kW} \tag{46}$$

The pump work is small compared with the turbine output because the working fluid is compressed in the liquid state.

### 1-2-3-3 Net Power Output

The net power output of the ORC is obtained by subtracting the pump power consumption from the turbine power output:

$$\dot{W_{net}} = \dot{W}_T - \dot{W}_P \tag{47}$$

Substituting the EES values:

$$\dot{W_{net}} = 51.01 - 1.468 \tag{48}$$

$$\dot{W_{net}} = 49.54\ \text{kW} \tag{49}$$

Therefore, the ORC with IHE produces a net power output of 49.54 kW for a geothermal fluid mass flow rate of 1 kg/s.

### 1-2-3-4 Heat Input to the ORC

The heat input to the working fluid in the evaporator is:

$$\dot{Q_{ORC}} = \dot{Q_{evap}} = \dot{m_{wf}}(h_4 - h_3) \tag{50}$$

From the EES output:

$$\dot{Q_{ORC}} = 359.6\ \text{kW} \tag{51}$$

This heat is supplied by the geothermal fluid in the evaporator.

### 1-2-3-5 Thermal Efficiency

The thermal efficiency of the ORC is defined as the ratio of net power output to the heat input received by the working fluid in the evaporator:

$$\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_{ORC}} \tag{52}$$

Using the EES results:

$$\eta_{th} = \frac{49.54}{359.6}$$
$$\eta_{th} = 0.1378 \tag{53}$$
$$\eta_{th} = 13.78\%$$

This efficiency represents the performance of the ORC itself based on the heat transferred to the organic working fluid. It should not be confused with the overall geothermal plant thermal efficiency, which is based on the total energy available in the geothermal fluid.

For comparison, the overall plant thermal efficiency calculated in EES is:

$$\eta_{plant} = 7.528\% \tag{54}$$

The value of 13.78% is used in the A-2 summary table because the requested performance table is for the ORC cycle performance.

## 1-2-4 Exergy Analysis

### 1-2-4-1 Specific Flow Exergy

The specific flow exergy is calculated as:

$$\psi = (h - h_0) - T_0(s - s_0) \tag{55}$$

where $h_0$ and $s_0$ are the enthalpy and entropy at the dead state. The dead-state conditions are:

$$T_0 = 25°C = 298.15\ \text{K} \tag{56}$$
$$P_0 = 100\ \text{kPa} \tag{57}$$

The exergy calculation must use the dead-state temperature in Kelvin.

### 1-2-4-2 Geothermal Exergy Input

The exergy input supplied by the geothermal fluid is:

$$\dot{E}_{geo,in} = \dot{m}_{geo}[(h_7 - h_0) - T_0(s_7 - s_0)] \tag{58}$$

Using the EES results:

$$\dot{E}_{geo,in} = 129.8\ \text{kW} \tag{59}$$

The geothermal fluid leaves the evaporator at state 8 with a remaining exergy rate of:

$$\dot{E}_{geo,out} = 31.64\ \text{kW} \tag{60}$$

This remaining exergy is associated with the reinjection stream.

#### 1-2-4-3 Exergy Efficiency

The exergy efficiency is defined as the ratio of net power output to the exergy input of the geothermal fluid:

$$\eta_{ex} = \frac{\dot{W}_{net}}{\dot{E}_{geo,in}} \tag{61}$$

Substituting the EES results:

$$\eta_{ex} = \frac{49.54}{129.8} = 0.3817 = 38.17\% \tag{62}$$

This means that 38.17% of the available geothermal exergy is converted into useful net power. The remaining part is destroyed due to irreversibilities or leaves the system with the reinjected geofluid and cooling air.

### 1-2-5 Component Exergy Destruction Equations

The total exergy destruction rate is calculated by summing the exergy destruction rates of the main components and the exergy losses associated with reinjection and cooling air.

#### 1-2-5-1 Pump

$$\dot{E}_{D,P} = T_0 \dot{m}_{wf}(s_2 - s_1) \tag{63}$$

From EES:

$$\dot{E}_{D,P} = 0.1374\ \text{kW} \tag{64}$$

The pump has a very small exergy destruction rate because the entropy increase during liquid compression is small.

#### 1-2-5-2 Turbine

$$\dot{E}_{D,T} = T_0 \dot{m}_{wf}(s_5 - s_4) \tag{65}$$

From EES:

$$\dot{E_{D,T}} = 11.39\ \text{kW} \tag{66}$$

The turbine exergy destruction is caused by irreversibilities during the expansion process.

#### 1-2-5-3 Internal Heat Exchanger

$$\dot{E_{D,IHE}} = T_0 \dot{m_{wf}}[(s_3 - s_2) + (s_6 - s_5)] \tag{67}$$

From EES:

$$\dot{E_{D,IHE}} = 0.6182\ \text{kW} \tag{68}$$

The IHE exergy destruction is relatively small. This indicates that the internal heat recovery process introduces limited irreversibility compared with the evaporator, condenser, and turbine.

#### 1-2-5-4 Evaporator

$$\dot{E_{D,evap}} = T_0\left[\dot{m_{wf}}(s_4 - s_3) + \dot{m_{geo}}(s_8 - s_7)\right] \tag{69}$$

From EES:

$$\dot{E_{D,evap}} = 21.36\ \text{kW} \tag{70}$$

The evaporator has one of the largest exergy destruction rates. This is expected because heat transfer occurs across a finite temperature difference between the geothermal fluid and the ORC working fluid.

#### 1-2-5-5 Condenser

$$\dot{E_{D,cond}} = T_0\left[\dot{m_{wf}}(s_1 - s_6) + \dot{m_{air}}(s_{10} - s_9)\right] \tag{71}$$

From EES:

$$\dot{E_{D,cond}} = 10.45\ \text{kW} \tag{72}$$

The condenser destroys exergy because heat is rejected from the working fluid to the cooling air at a finite temperature difference.

#### 1-2-5-6 Reinjection Exergy Loss

The exergy associated with the reinjected geothermal fluid is:

$$\dot{E_{D,rein}} = \dot{m_{geo}}[(h_8 - h_0) - T_0(s_8 - s_0)] \tag{73}$$

From EES:

$$\dot{E_{D,rein}} = 31.64\ \text{kW} \tag{74}$$

This is the largest contribution to the total exergy loss. It shows that the geothermal fluid still contains useful availability when it leaves the evaporator.

#### 1-2-5-7 Cooling Air Exergy Loss

The exergy carried by the cooling air leaving the condenser is:

$$\dot{E_{D,air}} = \dot{m_{air}}[(h_{10} - h_{air,0}) - T_0(s_{10} - s_{air,0})] \quad (75)$$

From EES:

$$\dot{E_{D,air}} = 4.655 \text{ kW} \quad (76)$$

This term represents the exergy carried away by the cooling air after absorbing heat in the condenser.

## 1-2-6 Total Exergy Destruction Rate

The total exergy destruction rate is calculated as:

$$\dot{E_{D,total}} = \dot{E_{D,P}} + \dot{E_{D,T}} + \dot{E_{D,IHE}} + \dot{E_{D,evap}} + \dot{E_{D,cond}} + \dot{E_{D,rein}} + \dot{E_{D,air}} \quad (77)$$

Substituting the EES values:

$$\dot{E_{D,total}} = 0.1374 + 11.39 + 0.6182 + 21.36 + 10.45 + 31.64 + 4.655 = 80.25\ kW \quad (78)$$

The same result is obtained from the overall exergy balance:

$$\dot{E_{D,total}} = \dot{E_{geo,in}} - \dot{W_{net}} = 129.8 - 49.54 = 80.25\ kW \quad (79)$$

The agreement between the component summation and the overall exergy balance confirms that the EES model is thermodynamically consistent.

*Table 9: Detailed Exergy Destruction Table of section A_2*

| Component / Loss Mechanism | Symbol | Value | Unit |
|---|---|---|---|
| **Pump** | $\dot{E}_{D,P}$ | 0.1374 | kW |
| **Turbine** | $\dot{E}_{D,T}$ | 11.39 | kW |
| **Internal heat exchanger** | $\dot{E}_{D,IHE}$ | 0.6182 | kW |
| **Evaporator** | $\dot{E}_{D,evap}$ | 21.36 | kW |
| **Condenser** | $\dot{E}_{D,cond}$ | 10.45 | kW |
| **Reinjection** | $\dot{E}_{D,rein}$ | 31.64 | kW |
| **Cooling air** | $\dot{E}_{D,air}$ | 4.655 | kW |
| **Total** | $\dot{E}_{D,total}$ | 80.25 | kW |

## 1-2-7 Discussion of Results

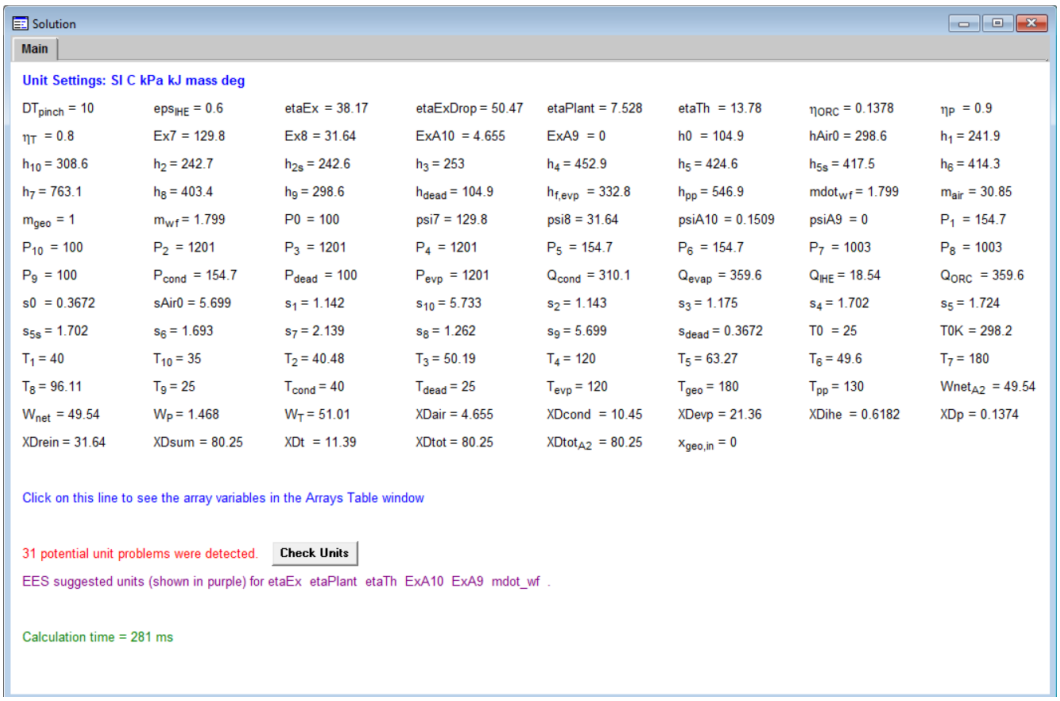



*Figure 3:EES simulation results for section A_2*

The EES simulation shows that the ORC with IHE produces 49.54 kW of net power for a geothermal fluid mass flow rate of 1 kg/s. The turbine produces 51.01 kW, while the pump consumes 1.468 kW. Therefore, the pump power consumption is small compared with the turbine output.

The working fluid mass flow rate is 1.799 kg/s. This value is determined from the evaporator pinch-point condition and the heat balance between the geothermal fluid and the ORC working fluid. The heat input to the ORC is 359.6 kW, giving a thermal efficiency of 13.78%. This value represents the thermal efficiency based on the heat absorbed by the working fluid in the evaporator.

The exergy input from the geothermal fluid is 129.8 kW. Since the net power output is 49.54 kW, the exergy efficiency is 38.17%. This result indicates that less than half of the geothermal exergy input is converted into useful power. The remaining exergy is destroyed inside the components or leaves the system with the reinjected geofluid and cooling air.

Among the calculated exergy destruction terms, the largest contribution is associated with reinjection, with a value of 31.64 kW. This means that the geothermal fluid still carries a considerable amount of useful energy after leaving the evaporator. The evaporator is the second major source of exergy destruction, with 21.36 kW, mainly due to heat transfer across a finite temperature difference. The turbine and condenser also contribute significantly, with exergy destruction rates of 11.39 kW and 10.45 kW, respectively.

The internal heat exchanger has a small exergy destruction rate of 0.6182 kW. This shows that the IHE improves heat recovery inside the cycle without introducing a large additional irreversibility. The pump has the smallest exergy destruction rate, equal to 0.1374 kW, because the entropy generation during liquid compression is very small.

Overall, the total exergy destruction rate is 80.25 kW. This value is consistent with the overall exergy balance, since the difference between the geothermal exergy input and the net power output is also 80.25 kW. Therefore, the performance summary obtained from the EES model is physically consistent

# 1-3 A_3: Exergy Destruction Rate Analysis of ORC-IHE Components

## 1-3-1 Objective

The objective of this section is to plot and analyze the exergy destruction rate of each component in the Organic Rankine Cycle with an Internal Heat Exchanger. The exergy destruction values were obtained from the EES simulation results of Part A-2 and were plotted as a bar chart using MATLAB.

The bar chart allows the main sources of irreversibility in the system to be identified. This is important because exergy destruction directly represents the loss of useful work potential inside each component. In Paper 1[2], the total exergy destruction rate of the binary geothermal cycle is also evaluated by summing the exergy destruction rates of the individual components, including the pump, evaporator, turbine, internal heat exchanger, condenser, reinjection stream, and cooling air stream.

## 1-3-2 Exergy Destruction Data

The exergy destruction rates obtained from the EES model are listed in Table 9.

*Table 10: Exergy destruction rates for section A_3 of project*

| Component / Loss Mechanism | Symbol | Exergy Destruction Rate | Unit |
|---|---|---|---|
| **Condenser** | $\dot{E}_{D,cond}$ | 10.45 | kW |
| **Cooling air** | $\dot{E}_{D,air}$ | 4.655 | kW |
| **Evaporator** | $\dot{E}_{D,evap}$ | 21.36 | kW |
| **Internal heat exchanger** | $\dot{E}_{D,IHE}$ | 0.6182 | kW |
| **Pump** | $\dot{E}_{D,P}$ | 0.1374 | kW |
| **Reinjection** | $\dot{E}_{D,rein}$ | 31.64 | kW |
| **Turbine** | $\dot{E}_{D,T}$ | 11.39 | kW |
| **Total** | $\dot{E}_{D,total}$ | 80.25 | kW |

## 1-3-3 Governing Equation

The total exergy destruction rate of the ORC-IHE system is calculated by summing the exergy destruction rates of all components:

$$\dot{E}_{D,total} = \dot{E}_{D,cond} + \dot{E}_{D,air} + \dot{E}_{D,evap} + \dot{E}_{D,IHE} + \dot{E}_{D,P} + \dot{E}_{D,rein} + \dot{E}_{D,T} \quad (80)$$

Substituting the EES results:

$$\dot{E}_{D,total} = 10.45 + 4.655 + 21.36 + 0.6182 + 0.1374 + 31.64 + 11.39 = 80.25\ kW \quad (81)$$

This value is consistent with the total exergy destruction rate obtained in Part A-2.

## 1-3-4 Exergy Destruction Share of Each Component

The percentage contribution of each component is calculated as:

$$Y_{D,i} = \frac{\dot{E_{D,i}}}{\dot{E_{D,total}}} \times 100 \tag{82}$$

*Table 11: Exergy Destruction Share of Each Component for section A_3 of project*

| Component / Loss Mechanism | Exergy Destruction Rate (kW) | Share of Total (%) |
|---|---|---|
| **Condenser** | 10.45 | 13.02 |
| **Cooling air** | 4.655 | 5.80 |
| **Evaporator** | 21.36 | 26.62 |
| **Internal heat exchanger** | 0.6182 | 0.77 |
| **Pump** | 0.1374 | 0.17 |
| **Reinjection** | 31.64 | 39.43 |
| **Turbine** | 11.39 | 14.19 |
| **Total** | 80.25 | 100.00 |

## 1-3-5 Figure Description

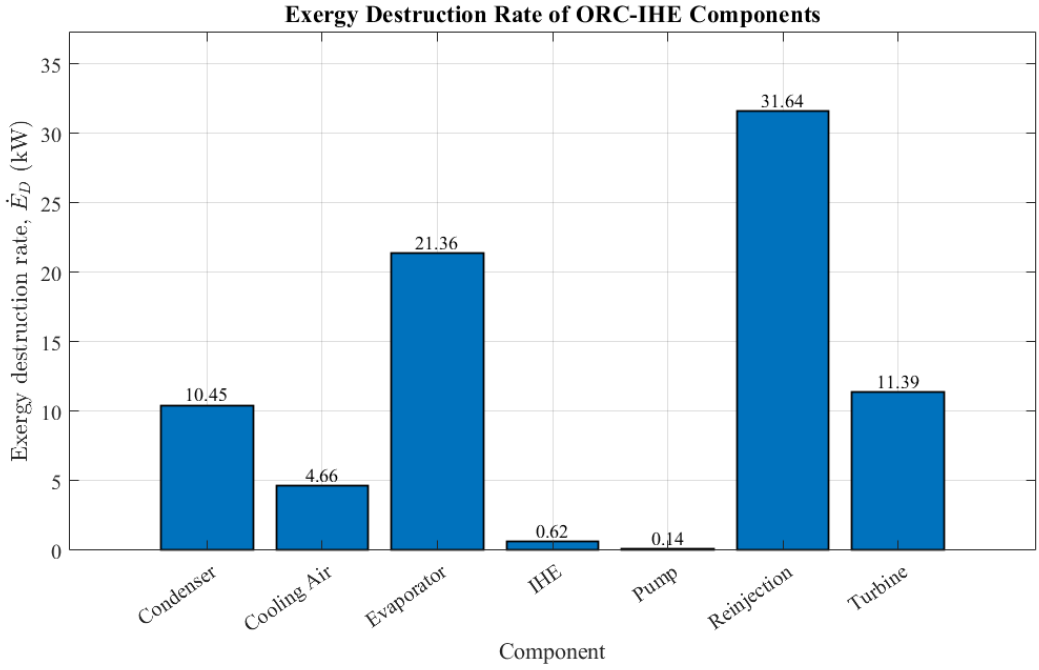



*Figure 4: Exergy destruction rate of each component in the ORC-IHE system for section A_3 of project*

The plotted results show that the reinjection stream has the highest exergy destruction rate, followed by the evaporator, turbine, condenser, cooling air, internal heat exchanger, and pump.

## 1-3-6 Discussion of Results

The bar chart shows that the largest exergy destruction is associated with the reinjection stream:

$$\dot{E}D, rein = 31.64\ kW \dot{E}_{D,rein} = 31.64\ \text{kW} \dot{E}D, rein = 31.64\ kW \quad (83)$$

This value corresponds to approximately 39.43% of the total exergy destruction rate. The high reinjection exergy loss means that the geothermal fluid still carries a significant amount of useful work potential after leaving the evaporator. Therefore, the reinjected geofluid is the dominant source of exergy loss in the system.

The evaporator is the second-largest source of irreversibility:

$$\dot{E}_{D,evap} = 21.36\ \text{kW} \quad (84)$$

This corresponds to approximately 26.62% of the total exergy destruction. The high exergy destruction in the evaporator is mainly caused by heat transfer across a finite temperature difference between the geothermal fluid and the ORC working fluid. Although the pinch-point temperature difference is set to $10°C$, irreversibility still occurs because the temperature profiles of the two fluids are not perfectly matched throughout the heat exchanger.

The turbine is the third major contributor to exergy destruction:

$$\dot{E}_{D,T} = 11.39\ \text{kW} \quad (85)$$

This represents approximately 14.19% of the total exergy destruction. The turbine irreversibility is caused by the non-isentropic expansion process. Since the turbine isentropic efficiency is 0.80, part of the available work potential is destroyed during expansion.

The condenser also has a significant exergy destruction rate:

$$\dot{E}_{D,cond} = 10.45\ \text{kW} \quad (86)$$

This corresponds to 13.02% of the total exergy destruction. The condenser irreversibility is mainly related to heat rejection from the working fluid to the cooling air at a finite temperature difference.

The cooling air stream carries away:

$$\dot{E}_{D,air} = 4.655\ \text{kW} \quad (87)$$

This accounts for 5.80% of the total exergy destruction. This term represents the useful work potential carried by the cooling air leaving the condenser at a temperature higher than the dead-state temperature.

The internal heat exchanger has a much smaller exergy destruction rate:

$$\dot{E}_{D,IHE} = 0.6182\ \text{kW} \quad (88)$$

This corresponds to only 0.77% of the total. Therefore, the IHE improves internal heat recovery without adding a large irreversibility penalty to the system.

The pump has the smallest exergy destruction rate:

$$\dot{E_{D,P}} = 0.1374\ \text{kW} \tag{89}$$

This is only 0.17% of the total exergy destruction. This small value is expected because the pump compresses the working fluid in the liquid phase, where the entropy generation and specific work input are relatively small.

## 1-3-7 Conclusion

The exergy destruction bar chart shows that the main irreversibilities of the ORC-IHE system occur in the reinjection stream and the evaporator. Together, these two terms account for approximately $39.43\% + 26.62\% = 66.05\%$ of the total exergy destruction. Therefore, the most effective improvement strategies should focus on reducing the exergy loss of the reinjected geothermal fluid and improving the thermal matching between the geothermal fluid and the ORC working fluid in the evaporator.

The turbine and condenser are also important sources of irreversibility, while the internal heat exchanger and pump have minor contributions. Overall, the total exergy destruction rate is:

$$\dot{E_{D,total}} = 80.25\ \text{kW} \tag{90}$$

which confirms that a considerable portion of the input geothermal exergy is lost due to component irreversibilities and unused exergy in the reinjected geothermal fluid.

# Part B: Heat Exchanger Area Calculation

## 2-1 Objective

The objective of this section is to calculate the required heat transfer area for the main heat exchangers in the ORC-IHE system. The considered components are the condenser, evaporator, and internal heat exchanger. The heat transfer rates and inlet/outlet temperatures were obtained from the EES simulation results of the previous sections, while the overall heat transfer coefficients were taken from Table 2.

The heat exchanger areas were calculated using the logarithmic mean temperature difference method. A counter-flow arrangement was assumed for all heat exchangers.

## 2-2 Governing Equations

The heat transfer rate in a heat exchanger is expressed as:

$$\dot{Q} = UA\Delta T_{LMTD} \tag{91}$$

Therefore, the required heat transfer area is:

$$A = \frac{\dot{Q}}{U\Delta T_{LMTD}} \tag{92}$$

The logarithmic mean temperature difference is calculated from:

$$\Delta T_{LMTD} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)} \tag{93}$$

where $\Delta T_1$ and $\Delta T_2$ are the two terminal temperature differences of the heat exchanger.

For all calculations, the heat transfer rate was converted from kW to W, $1kW = 1000\,W$.

This conversion is necessary because the heat transfer coefficient U is given in $\mathrm{W/m^2.K}$.

## 2-3 Data Used in the Calculations

The following data were obtained from the EES simulation:

*Table 12: Data Used in the Calculations for part B of project*

| Parameter | Value | Unit |
|---|---|---|
| $\mathbf{Q_{cond}}$ | 310.1 | kW |
| $\mathbf{Q_{evap}}$ | 359.6 | kW |
| $\mathbf{Q_{IHE}}$ | 18.54 | kW |
| $\mathbf{T_1}$ | 40.00 | °C |
| $\mathbf{T_2}$ | 40.48 | °C |
| $\mathbf{T_3}$ | 50.19 | °C |
| $\mathbf{T_4}$ | 120.00 | °C |
| $\mathbf{T_5}$ | 63.27 | °C |
| $\boldsymbol{T_6}$ | 49.60 | °C |

| | | |
|---|---|---|
| $T_7$ | 180.00 | $°C$ |
| $T_8$ | 96.11 | $°C$ |
| $T_9$ | 25.00 | $°C$ |
| $T_{10}$ | 35.00 | $°C$ |

*Table 13: The overall heat transfer coefficients of section B of project*

| Component | U | Unit |
|---|---|---|
| **Condenser** | 250 | $W/m^2.K$ |
| **Internal Heat Exchanger** | 300 | $W/m^2.K$ |
| **Evaporator** | 850 | $W/m^2.K$ |

For the IHE, the value $U = 300\ W/m^2.K$ was used because Table 2 lists this value for the heat exchanger. The pre-heater value was not used because the required components in this section are condenser, evaporator, and IHE.

## 2-4 Condenser Area Calculation

In the condenser, the ORC working fluid rejects heat to the cooling air. The hot stream is the R123 working fluid, and the cold stream is air.

The temperatures are:

$$T_{h,in} = T_6 = 49.60°C \quad (94)$$

$$T_{h,out} = T_1 = 40.00°C \quad (95)$$

$$T_{c,in} = T_9 = 25.00°C \quad (96)$$

$$T_{c,out} = T_{10} = 35.00° \quad (97)$$

For a counter-flow heat exchanger:

$$\Delta T_1 = T_{h,in} - T_{c,out} = 49.60 - 35.00 = 14.60\ K \quad (98)$$

$$\Delta T_1 = 49.60 - 35.00 = 14.60\ K$$
$$\Delta T_2 = 40.00 - 25.00 = 15.00\ K \quad (99)$$

Therefore:

$$\Delta T_{LMTD,cond} = \frac{14.60 - 15.00}{\ln\left(\frac{14.60}{15.00}\right)} = 14.80\ K \quad (100)$$

The condenser area is:

$$A_{cond} = \frac{\dot{Q}_{cond}}{U_{cond}\Delta T_{LMTD,cond}}$$

$$A_{cond} = \frac{\dot{Q}_{cond}}{U_{cond}\Delta T_{LMTD,cond}} \quad ()$$

$$A_{cond} = \frac{310.1 \times 1000}{250 \times 14.80} = 83.82\ m^2$$

## 2-5 Evaporator Area Calculation

In the evaporator, heat is transferred from the geothermal fluid to the ORC working fluid. The hot stream is the geothermal fluid, and the cold stream is R123.

The temperatures are:

$$T_{h,in} = T_7 = 180.00°C \quad (101)$$

$$T_{h,out} = T_8 = 96.11°C \quad (102)$$

$$T_{c,in} = T_3 = 50.19°C \quad (103)$$

$$T_{c,out} = T_4 = 120.00°C \quad (104)$$

For counter-flow operation:

$$\Delta T_1 = T_{h,in} - T_{c,out} = 180.00 - 120.00 = 60.00\ K \quad (105)$$

$$\Delta T_2 = T_{h,out} - T_{c,in} = 96.11 - 50.19 = 45.92\ K \quad (106)$$

Thus:

$$\Delta T_{LMTD,evap} = \frac{60.00 - 45.92}{\ln\left(\frac{60.00}{45.92}\right)} = 52.65\ K \quad (107)$$

The evaporator area is:

$$A_{evap} = \frac{\dot{Q}_{evap}}{U_{evap}\Delta T_{LMTD,evap}}$$

$$A_{evap} = \frac{359.6 \times 1000}{850 \times 52.65} = 8.04\ \text{m}^2 \quad (108)$$

## 2-6 Internal Heat Exchanger Area Calculation

In the internal heat exchanger, the hot turbine exhaust transfers heat to the pressurized working fluid leaving the pump. Both streams are R123.

The hot-side temperatures are:

$$T_{h,in} = T_5 = 63.27°C \quad (109)$$

$$T_{h,out} = T_6 = 49.60°C \quad (110)$$

The cold-side temperatures are:

$$T_{c,in} = T_2 = 40.48°C \quad (111)$$

$$T_{c,out} = T_3 = 50.19°C \quad (112)$$

For counter-flow arrangement:

$$\Delta T_1 = T_{h,in} - T_{c,out} = 63.27 - 50.19 = 13.08\ K \quad (113)$$

$$\Delta T_2 = T_{h,out} - T_{c,in} = 49.60 - 40.48 = 9.12\ K \quad (114)$$

Therefore:

$$\Delta T_{LMTD,IHE} = \frac{13.08 - 9.12}{\ln\left(\frac{13.08}{9.12}\right)} = 10.98\ K \quad (115)$$

The IHE area is:

$$A_{IHE} = \frac{\dot{Q_{IHE}}}{U_{IHE}\Delta T_{LMTD,IHE}}$$
$$A_{IHE} = \frac{18.54 \times 1000}{300 \times 10.98} = 5.63\ \text{m}^2 \quad (116)$$

## 2-7 Heat Exchanger Area Results

*Table 14: Heat Exchanger Area Results for section B of project*

| Heat Exchanger | $\dot{Q}$ (kW) | U ($W/m^2.K$) | $\Delta T_1$ (K) | $\Delta T_2$ (K) | $\Delta T_{LMTD}$ (K) | Area ($m^2$) |
|---|---|---|---|---|---|---|
| **Condenser** | 310.1 | 250 | 14.60 | 15.00 | 14.80 | 83.82 |
| **Evaporator** | 359.6 | 850 | 60.00 | 45.92 | 52.65 | 8.04 |
| **Internal Heat Exchanger** | 18.54 | 300 | 13.08 | 9.12 | 10.98 | 5.63 |

## 2-8 Discussion

The condenser has the largest required heat transfer area:

$$A_{cond} = 83.82\ \text{m}^2 \quad (117)$$

This large area is mainly due to the low overall heat transfer coefficient of the condenser and the small logarithmic mean temperature difference. Although the condenser heat rejection rate is slightly lower than the evaporator heat transfer rate, the small U value and low temperature difference significantly increase the required surface area.

The evaporator has a calculated area of:

$$A_{evap} = 8.04\ \text{m}^2 \quad (118)$$

Although the evaporator has the highest heat transfer rate among the three components, its required area is much smaller than the condenser area. This is because the evaporator has a much larger overall heat transfer coefficient and a higher logarithmic mean temperature difference.

The internal heat exchanger has the smallest heat transfer rate:

$$\dot{Q_{IHE}} = 18.54\ \text{kW} \quad (119)$$

Its required area is:

$$A_{IHE} = 5.63\ \text{m}^2 \quad (120)$$

This value is reasonable because the IHE transfers a relatively small amount of heat compared with the condenser and evaporator. However, its LMTD is also relatively small, which prevents the area from becoming negligible.

Overall, the results show that the condenser dominates the total heat exchanger area requirement of the system. This indicates that condenser sizing is an important design consideration in the ORC-IHE system.

# Part C: Economic Analysis

## 3-1 Objective

The objective of this section is to perform a thermo-economic analysis of the ORC with Internal Heat Exchanger simulated in Part A. The economic model is based on the cost-rate balance, component cost functions, and economic parameters provided in Paper 2[1]. In that paper, the economic model includes the fuel cost rate, capital investment cost, operation and maintenance cost, and capital recovery factor.

The main required output of this section is the total product cost rate of the system $\dot{C_{pro}}$. In addition, the specific production cost is also reported as $c_{pro}$, because Paper 2[1] commonly reports production cost in $\frac{\$}{GJ}/$.

## 3-2 Economic Model

The total product cost rate is calculated as the sum of the fuel cost rate and the total equipment cost rate:

$$\dot{C_{pro}} = \dot{C_{fuel}} + \dot{Z_{tot}} \tag{121}$$

The fuel cost rate is calculated from the geofluid exergy input:

$$\dot{C_{fuel}} = c_{gf}\dot{E_{x,gf}} \tag{122}$$

Since $c_{gf}$ is given in $\$/GJ$, the exergy rate in kW must be converted to GJ/s:

$$\dot{C_{fuel}} = c_{gf}\frac{\dot{E_{x,gf}}}{10^6} \tag{123}$$

The equipment cost rate is obtained by annualizing the capital investment cost of each component:

$$\dot{Z_k} = \frac{Z_k\,\varphi\,CRF}{h \times 3600} \tag{124}$$

where $Z_k$ is the capital investment cost of component $k, \varphi$ is the maintenance factor,h is the annual operating hours, and $CRF$ is the capital recovery factor. Paper 2[1] defines the cost-rate balance, the fuel cost rate, the operation and maintenance cost-rate expression, and the CRF relation in its economic analysis section.

The capital recovery factor is:

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1} \tag{125}$$

where iii is the annual interest rate and nnn is the system lifetime.

## 3-3 Economic Input Data

The economic parameters used in this analysis are taken from Paper 2[1]. In Table 3 of Paper 2[1], the annual operating hours, unit cost of geofluid exergy, system lifetime, annual interest rate, and maintenance factor are given as $7000\ h/year$, $1.3\ \$/GJ$, $20\ years$, 14%, and 1.06, respectively.

*Table 15: Economic Input Data fot section C of project*

| Parameter | Symbol | Value | Unit |
|---|---|---|---|
| **Annual operating hours** | h | 7000 | h/year |
| **Unit cost of geofluid exergy** | $c_{gf}$ | 1.3 | $/GJ |
| **System lifetime** | n | 20 | year |
| **Annual interest rate** | i | 0.14 | – |
| **Maintenance factor** | φ | 1.06 | – |

The calculated capital recovery factor is:

$$CRF = 0.151 \tag{126}$$

## 3-4 Thermodynamic and Design Inputs from Previous Sections

The following values were obtained from the EES simulation and the heat exchanger area calculation.

*Table 16: Thermodynamic and Design Inputs for section C of project*

| Parameter | Symbol | Value | Unit |
|---|---|---|---|
| **Turbine power output** | $\dot{W}_T$ | 51.01 | kW |
| **Pump power consumption** | $\dot{W}_P$ | 1.468 | kW |
| **Net power output** | $\dot{W}_{net}$ | 49.54 | kW |
| **Geofluid exergy input** | $\dot{E}_{x,gf}$ | 129.8 | kW |
| **Condenser area** | $A_{cond}$ | 83.81 | $m^2$ |
| **Evaporator area** | $A_{evap}$ | 8.036 | $m^2$ |
| **Internal heat exchanger area** | $A_{IHE}$ | 5.629 | $m^2$ |

## 3-5 Component Capital Cost Functions

The capital investment cost functions used for the turbine, pump, evaporator, condenser, and internal heat exchanger are taken from Table 2 of Paper 2[1]. The paper gives the turbine cost as a function of turbine power, the pump cost as a function of pump power, the evaporator and condenser cost as a function of heat transfer area, and the internal heat exchanger cost as a function of heat exchanger area.

### 3-5-1 Turbine

$$Z_{tur} = 4405\dot{W_T^{0.89}} \tag{127}$$

### 3-5-2 Pump

$$Z_{pump} = 1120\dot{W_P^{0.8}} \tag{128}$$

### 3-5-3 Condenser

$$Z_{cond} = 1397A_{cond}^{0.89} \tag{129}$$

### 3-5-4 Evaporator

$$Z_{evap} = 1397 A_{evap}^{0.89} \tag{130}$$

### 3-5-5 Internal Heat Exchanger

$$Z_{IHE} = 2143 A_{IHE}^{0.514} \tag{131}$$

## 3-6 Capital Investment Cost Results

Using the cost functions above, the capital investment cost of each component was calculated.

*Table 17: Capital Investment Cost Results for section C of project*

| Component | Symbol | Value | Unit |
|---|---|---|---|
| **Turbine** | $Z_{tur}$ | 145805 | $ |
| **Pump** | $Z_{pump}$ | 1523 | $ |
| **Condenser** | $Z_{cond}$ | 71935 | $ |
| **Evaporator** | $Z_{evap}$ | 8927 | $ |
| **Internal heat exchanger** | $Z_{IHE}$ | 5209 | $ |
| **Total capital investment** | $Z_{tot}$ | **233398** | $ |

The turbine has the largest capital investment cost, followed by the condenser. The high turbine cost is due to the dependence of its cost function on power output. The condenser cost is also significant because the condenser has the largest heat transfer area among the heat exchangers.

## 3-7 Equipment Cost Rate Calculation

The cost rate of each component is calculated from:

$$\dot{Z_k} = \frac{Z_k \, \varphi \, CRF}{h \times 3600} \tag{132}$$

For the turbine:

$$\dot{Z_{tur}} = \frac{145805 \times 1.06 \times 0.151}{7000 \times 3600}$$
$$\dot{Z_{tur}} = 0.000926 \, \$/s \tag{133}$$

The same method was applied to all components.

*Table 18: Equipment Cost Rate Calculation for section C of project*

| Component | Symbol | Value | Unit |
|---|---|---|---|
| **Turbine** | $\dot{Z}_{tur}$ | 0.000926 | $/s |
| **Pump** | $\dot{Z}_{pump}$ | 0.000009671 | $/s |
| **Condenser** | $\dot{Z}_{cond}$ | 0.000456 | $/s |
| **Evaporator** | $\dot{Z}_{evap}$ | 0.00005669 | $/s |
| **Internal heat exchanger** | $\dot{Z}_{IHE}$ | 0.00003308 | $/s |
| **Total equipment cost rate** | $\dot{Z}_{tot}$ | **0.001482** | $/s |

Thus:

$$\dot{Z_{tot}} = 0.001482\ \$/s \tag{134}$$

## 3-8 Fuel Cost Rate

The geofluid exergy input is:

$$\dot{E_{x,gf}} = 129.8\ \text{kW} \tag{135}$$

The unit cost of geofluid exergy is:

$$\dot{C_{fg}} = 1.3\ \$/GJ \tag{136}$$

Therefore:

$$\dot{C_{fuel}} = 1.3\frac{129.8}{10^6}$$
$$\dot{C_{fuel}} = 0.0001687\ \$/s \tag{137}$$

The fuel cost rate is lower than the equipment cost rate, which means that the capital-related cost dominates the total product cost rate in this small-scale system.

## 3-9 Total Product Cost Rate

The total product cost rate is:

$$\dot{C_{pro}} = \dot{C_{fuel}} + \dot{Z_{tot}} \tag{138}$$

Substituting the calculated values:

$$\dot{C_{pro}} = 0.0001687 + 0.001482 = 0.001651\ \$/s \tag{139}$$

In hourly form:

$$\dot{C_{pro,h}} = 0.001651 \times 3600 = 5.944\ \$/h \tag{140}$$

Therefore, the calculated total product cost rate is:

$$\dot{C_{pro}} = 0.001651\ \$/s \tag{141}$$

or:

$$\boxed{\dot{C_{pro}} = 5.944\ \$/h} \tag{142}$$

## 3-10 Specific Production Cost

Although the requested value is the total product cost rate, the specific production cost is also useful for comparison with Paper 2[1], because Paper 2[1] reports production cost in $\$/GJ$. The paper presents economic optimization results using production cost as one of the objective functions and reports optimized values of $c_{pro}$ in $\$/GJ$.

The specific production cost is calculated as:

$$c_{pro} = \frac{\dot{C_{pro}}}{\dot{W_{net}}} \tag{143}$$

To express it in $\$/GJ$:

$$c_{pro} = \frac{\dot{C_{pro}}}{\dot{W_{net}}} \times 10^6 = \frac{0.001651}{49.54} \times 10^6 = 33.33\ \$/GJ \tag{144}$$

In $\$/kWh$:

$$c_{pro} = 33.33 \times 0.0036 = 0.120\ \$/kWh \tag{145}$$

Thus:

$$c_{pro} = 33.33\ \$/GJ\ =\ 0.120\ \$/kWh \tag{146}$$

## 3-11 Final Economic Results Table




*Figure 5:EES simulation results for section C of project*

*Table 19: Final Economic Results Table for section C of project*

| Parameter | Symbol | Value | Unit |
|---|---|---|---|
| **Capital recovery factor** | CRF | 0.151 | – |
| **Total capital investment** | $Z_{tot}$ | 233398 | $ |
| **Total equipment cost rate** | $\dot{Z}_{tot}$ | 0.001482 | $/s |
| **Fuel cost rate** | $\dot{C}_{fuel}$ | 0.0001687 | $/s |
| **Total product cost rate** | $\dot{C}_{pro}$ | 0.001651 | $/s |
| **Total product cost rate** | $\dot{C}_{pro,h}$ | 5.944 | $/h |
| **Specific production cost** | $c_{pro}$ | 33.33 | $/GJ |
| **Specific production cost** | $c_{pro}$ | 0.120 | $/kWh |

## 3-12 Discussion

The thermo-economic results show that the total capital investment of the simulated ORC-IHE system is 233398 $. The turbine is the most expensive component, with a capital cost of 145805 $. The condenser is the second most expensive component, with a capital cost of 71935 $, mainly because it has the largest heat transfer area.

The total equipment cost rate is:

$$\dot{Z_{tot}} = 0.001482\ \$/s \tag{147}$$

while the fuel cost rate associated with the geothermal exergy input is:

$$\dot{C_{fuel}} = 0.0001687\ \$/s \tag{148}$$

Therefore, the total product cost rate is mainly controlled by the equipment cost rate rather than the geofluid cost rate. This is reasonable because the assigned unit cost of geofluid exergy is relatively low.

The final total product cost rate is:

$$\dot{C_{pro}} = 0.001651\ \$/s \tag{149}$$

which is equivalent to:

$$5.944\ \$/h \tag{150}$$

The specific production cost is:

$$c_{pro} = 33.33\ \$/GJ \tag{151}$$

or:

$$c_{pro} = 0.120\ \$/kWh \tag{152}$$

This value is higher than the optimized values reported in Paper 2[1] for large-scale IHE-ORC systems. The difference is expected because the present cycle was simulated at a smaller scale with $\dot{m_{geo}} = 1\ \mathrm{kg/s}$ and $\dot{W_{net}} = 49.54\ \mathrm{kW}$. Small-scale ORC systems generally have higher specific production costs because equipment investment does not decrease linearly with system size.

## 3-13 Conclusion

The economic analysis of the ORC-IHE system was performed using the cost correlations and economic parameters provided in Paper 2[1]. The calculated total product cost rate was:

$$\boxed{\dot{C_{pro}} = 0.001651\ \$/s} \tag{153}$$

or:

$$\boxed{\dot{C_{pro}} = 5.944\ \$/h} \tag{154}$$

The corresponding specific production cost was:

$$c_{pro} = 33.33\ \$/GJ \tag{155}$$

or:

$$\boxed{c_{pro} = 0.120\ \$/kWh} \tag{156}$$

The results show that the turbine and condenser are the dominant contributors to the capital investment cost. The total equipment cost rate is significantly larger than the fuel cost rate, indicating that the economics of the simulated ORC-IHE system are mainly controlled by equipment sizing and capital investment.

# Part D: Parametric Study

## 4-1 D-1) Effect of Pinch Point Temperature Difference on ORC-IHE Performance

### 4-1-1 Objective

The objective of this section is to investigate the effect of the evaporator pinch point temperature difference on the performance of the ORC with Internal Heat Exchanger. The pinch point temperature difference, $\Delta T_{pinch}$, was varied from $5°C\ to\ 20°C$, while all other input parameters were kept constant.

The investigated parameters are:

$$A_{evap} \dot{m_{wf}} \eta_{ex} \tag{157}$$

where $A_{evap}$ is the evaporator heat transfer area, $\dot{m_{wf}}$ is the working fluid mass flow rate, and $\eta_{ex}$ is the exergy efficiency.

In Paper 2[1], the heat exchange process in the evaporator is also modeled using a fixed pinch point temperature difference. The pinch point is defined as the temperature difference between the working fluid vaporization temperature and the geothermal brine line in the temperature-entropy diagram.

## 4-2 Governing Equations

The pinch point temperature is defined as:

$$T_{pp} = T_{evp} + \Delta T_{pinch} \tag{158}$$

The working fluid mass flow rate is calculated from the evaporator pinch point energy balance:

$$\dot{m_{wf}}(h_4 - h_{f,evp}) = \dot{m_{geo}}(h_7 - h_{pp}) \tag{159}$$

where $h_{pp}$ is the geofluid enthalpy at the pinch point temperature and $h_{f,evp}$ is the saturated liquid enthalpy of R123 at the evaporator temperature.

The evaporator heat transfer rate is:

$$\dot{Q_{evap}} = \dot{m_{wf}}(h_4 - h_3) \tag{160}$$

The evaporator area is calculated using the LMTD method:

$$\dot{Q_{evap}} = U_{evap} A_{evap} \Delta T_{LMTD} \tag{161}$$

Therefore:

$$A_{evap} = \frac{\dot{Q_{evap}}}{U_{evap}\Delta T_{LMTD}} \tag{162}$$

The logarithmic mean temperature difference is:

$$\Delta T_{LMTD} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)} \tag{163}$$

In Paper 2[1], the heat exchanger area is calculated using the same general relation, $\dot{Q} = UAF\Delta T_{LMTD}$, and for counter-flow heat exchangers the correction factor is taken as F=1.

The exergy efficiency is defined as:

$$\eta_{ex} = \frac{\dot{W_{net}}}{\dot{E_{x,gf}}} \times 100 \tag{164}$$

where $\dot{E_{x,gf}}$ is the geofluid exergy input.

## 4-2-1 EES Simulation results

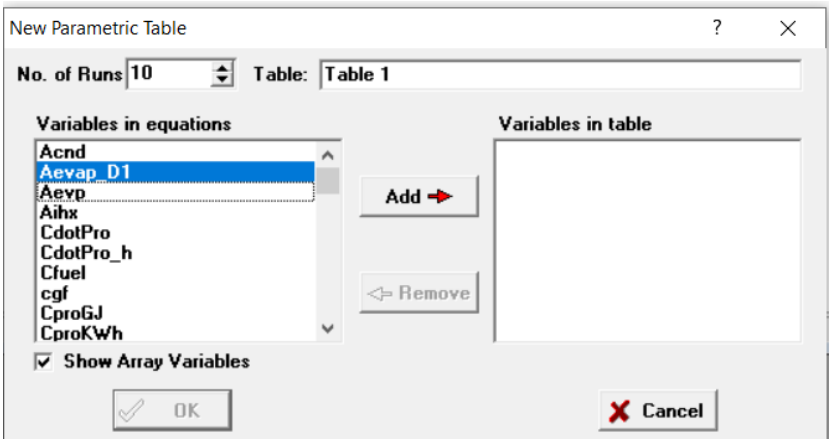



*Figure 6:Choosing reauired items for table calculation in EES software for section D_1 of project*

Parametric Table

Table 1

| 1..7 | $Aevap_{D1}$ | $DT_{pinch}$ | DTe2 | $mwf_{D1}$ | $etaEx_{D1}$ | $Wnet_{D1}$ | $Qevap_{D1}$ | $Tgeo_{out,D1}$ | $Cpro_{D1}$ |
|---|---|---|---|---|---|---|---|---|---|
| Run 1 | 9.707 | 5 | 37.5 | 1.976 | 41.93 | 54.42 | 395 | 87.69 | 32.79 |
| Run 2 | 8.825 | 7.5 | 41.71 | 1.887 | 40.05 | 51.98 | 377.3 | 91.9 | 33.04 |
| Run 3 | 8.036 | 10 | 45.92 | 1.799 | 38.17 | 49.54 | 359.6 | 96.11 | 33.33 |
| Run 4 | 7.324 | 12.5 | 50.13 | 1.71 | 36.29 | 47.1 | 341.9 | 100.3 | 33.64 |
| Run 5 | 6.676 | 15 | 54.34 | 1.621 | 34.4 | 44.66 | 324.1 | 104.5 | 33.99 |
| Run 6 | 6.081 | 17.5 | 58.55 | 1.532 | 32.52 | 42.21 | 306.4 | 108.7 | 34.38 |
| Run 7 | 5.532 | 20 | 62.75 | 1.443 | 30.63 | 39.76 | 288.6 | 112.9 | 34.81 |

*Figure 7: Calculated table by EES for section D_1 of project*

## 4-2-2 Parametric Results

The EES parametric table results are summarized below.

*Table 20: EES parametric table results for section D_1 of project*

| $\Delta T_{pinch}$ $(°C)$ | $A_{evap}$ $(m^2)$ | $\dot{m}_{wf}$ $(kg/s)$ | $\eta_{ex}$ (%) | $\dot{W}_{net}$ $(kW)$ | $\dot{Q}_{evap}$ $(kW)$ | $T_{geo,out}$ $(°C)$ |
|---|---|---|---|---|---|---|
| **5.0** | 9.707 | 1.976 | 41.93 | 54.42 | 395.0 | 87.69 |
| **7.5** | 8.825 | 1.887 | 40.05 | 51.98 | 377.3 | 91.90 |
| **10.0** | 8.036 | 1.799 | 38.17 | 49.54 | 359.6 | 96.11 |
| **12.5** | 7.324 | 1.710 | 36.29 | 47.10 | 341.9 | 100.3 |
| **15.0** | 6.676 | 1.621 | 34.40 | 44.66 | 324.1 | 104.5 |
| **17.5** | 6.081 | 1.532 | 32.52 | 42.21 | 306.4 | 108.7 |
| **20.0** | 5.532 | 1.443 | 30.63 | 39.76 | 288.6 | 112.9 |

## 4-2-3 Effect on Evaporator Area

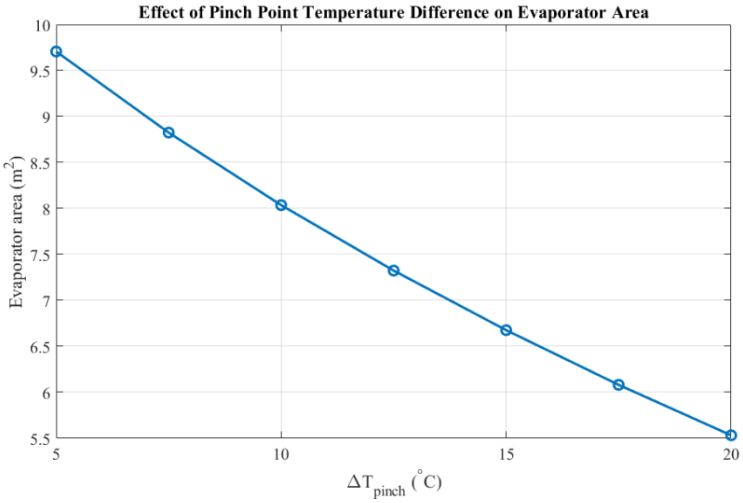



*Figure 8: effect of $\Delta T_{pinch}$ on the evaporator area*

The evaporator area decreases continuously as the pinch point temperature difference increases.

- At $\Delta T_{pinch} = 5°C$, the required evaporator area is $A_{evap} = 9.707\ m^2$.
- At $\Delta T_{pinch} = 20°C$, it decreases to $A_{evap} = 5.532\ m^2$.

This corresponds to a reduction of about:

$$\frac{9.707 - 5.532}{9.707} \times 100 = 43.0\% \tag{165}$$

The reason is that a larger pinch point temperature difference reduces the amount of heat recovered from the geothermal fluid. As $\Delta T_{pinch}$ increases, the geofluid outlet temperature rises from $87.69°C$ to $112.9°C$, meaning that less heat is extracted in the evaporator. Consequently, the

evaporator heat duty decreases from 395.0 kW to 288.6 kW. Since the heat duty decreases significantly, the required evaporator area also decreases.

### 4-2-4 Effect on Working Fluid Mass Flow Rate

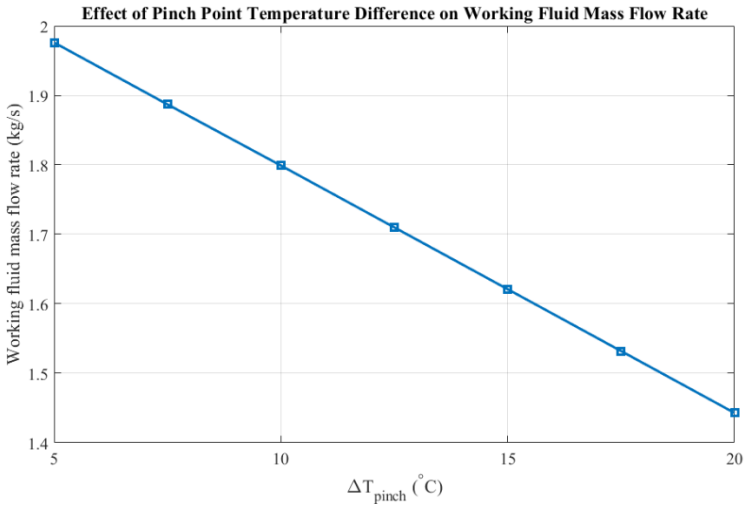



*Figure 9: Effect of $\Delta T_{pinch}$ on the working fluid mass flow rate*

The working fluid mass flow rate decreases almost linearly as the pinch point temperature difference increases.

- At $\Delta T_{pinch} = 5°C$, the working fluid mass flow rate is $\dot{m_{wf}} = 1.976$ kg/s.
- At $\Delta T_{pinch} = 20°C$, it decreases to $\dot{m_{wf}} = 1.443\ kg/s$

The reduction is approximately:

$$\frac{1.976 - 1.443}{1.976} \times 100 = 27.0\% \tag{166}$$

This behavior is physically expected. Increasing $\Delta T_{pinch}$ increases the pinch point temperature $T_{pp}$. As a result, the available enthalpy drop of the geothermal fluid before the pinch point decreases. Therefore, less heat is available for evaporating the working fluid, and the working fluid mass flow rate decreases. Paper 2[1] reports the same general trend: higher pinch point temperature differences reduce heat transfer and decrease the working fluid mass flow rate.

### 4-2-5 Effect on Exergy Efficiency

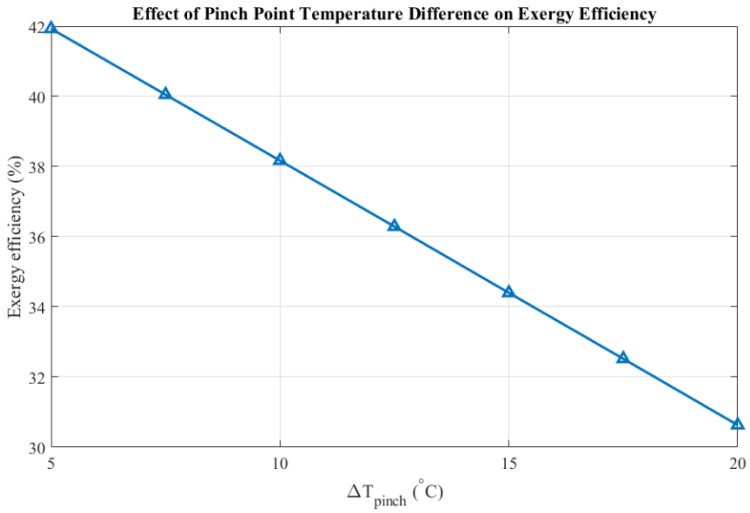



*Figure 10: Effect of $\Delta T_{pinch}$ on the exergy efficiency*

The exergy efficiency decreases continuously as the pinch point temperature difference increases. At $\Delta T_{pinch} = 5°C$, the exergy efficiency is $\eta_{ex} = 41.93\%$ At $\Delta T_{pinch} = 20°C$, it decreases to $\eta_{ex} = 30.63\%$. This corresponds to a decrease of $41.93 - 30.63 = 11.30$ percentage points.

The decrease in exergy efficiency occurs because the geofluid inlet exergy remains approximately constant, while the net power output decreases from $54.42\ kW$ to $39.76\ kW$. Since:

$$\eta_{ex} = \frac{\dot{W_{net}}}{\dot{E_{x,gf}}} \times 100 \tag{167}$$

a lower net power output directly reduces the exergy efficiency.

### 4-2-6 Discussion

The parametric results show a clear trade-off between heat exchanger size and thermodynamic performance. Increasing the pinch point temperature difference reduces the required evaporator area, which may be favorable from the viewpoint of equipment size and capital cost. However, this improvement comes at the cost of lower heat recovery from the geothermal fluid.

At higher values of $\Delta T_{pinch}$, the geothermal fluid leaves the evaporator at a higher temperature. This means that a larger portion of the geothermal energy remains unused in the outlet geofluid stream. As a result, the working fluid mass flow rate decreases, the evaporator heat duty decreases, and the turbine produces less power.

The reduction in working fluid mass flow rate also reduces the net power output of the ORC-IHE system. Since the exergy input from the geothermal fluid is almost unchanged, the lower net power output leads to a lower exergy efficiency.

Therefore, selecting a very high pinch point temperature difference is not favorable from a thermodynamic viewpoint. On the other hand, selecting a very low pinch point temperature

difference requires a larger evaporator area. Thus, the pinch point temperature difference should be selected by considering both thermodynamic performance and heat exchanger cost.

## 4-3 Conclusion

In this section, the pinch point temperature difference was varied from $5°C$ to $20°C$. The results show that increasing $\Delta T_{pinch}$ decreases the evaporator area, working fluid mass flow rate, and exergy efficiency.

- The evaporator area decreases from $29.707\ m^2$ to $25.532\ m^2$.
- The working fluid mass flow rate decreases from $1.976\ kg/s$ to $1.443\ kg/s$.
- The exergy efficiency decreases from $41.93\%$ to $30.63\%$.

Therefore, a larger pinch point temperature difference reduces the evaporator size but weakens the cycle performance. A lower pinch point temperature difference improves heat recovery and exergy efficiency, but it requires a larger evaporator area. The results demonstrate that the pinch point temperature difference is a key design parameter that must be optimized by considering both thermal performance and equipment size

## 4-4 D-2) Effect of Evaporator Temperature on Thermal and Exergy Efficiencies

### 4-4-1 Objective

The objective of this section is to investigate the effect of evaporator temperature on the thermal and exergy performance of the ORC with Internal Heat Exchanger. In this parametric study, the evaporator temperature was varied from $120°C \leq T_{evp} \leq 140°C$ while the condenser temperature, geothermal inlet temperature, pinch point temperature difference, turbine efficiency, pump efficiency, and IHE effectiveness were kept constant.

The investigated output parameters are $\eta_{th}$and $\eta_{ex}$.

where $\eta_{th}$ is the thermal efficiency and $\eta_{ex}$ is the exergy efficiency.

Evaporator pressure is one of the key design parameters in ORC systems. Paper 2[1] also studies the effect of evaporator pressure on net power output, thermodynamic efficiencies, and production cost for both ORC and IHE-ORC systems. In the present model, changing TevpT_{evp}Tevp directly changes the saturation pressure of R123 in the evaporator.

### 4-4-2 Governing Equations

The evaporator pressure is obtained from the saturation pressure of R123 at the evaporator temperature:

$$P_{evp} = P_{sat,R123}(T_{evp}) \tag{168}$$

Therefore, increasing $T_{evp}$ increases $P_{evp}$. According to the EES results, the evaporator pressure increases from $P_{evp} = 1201$ kPa at $T_{evp} = 120°C$, to $P_{evp} = 1760$ kPa at $T_{evp} = 140°C$.

The thermal efficiency is calculated as:

$$\eta_{th} = \frac{\dot{W_{net}}}{\dot{Q_{ORC}}} \times 100 \tag{169}$$

where:

$$\dot{Q_{ORC}} = \dot{Q_{evap}} = \dot{m_{wf}}(h_4 - h_3) \tag{170}$$

The exergy efficiency is defined as:

$$\eta_{ex} = \frac{\dot{W_{net}}}{\dot{E_{x,gf}}} \times 100 \tag{171}$$

The specific flow exergy relation used in the model is:

$$ex = (h - h_0) - T_0(s - s_0) \tag{172}$$

and the exergy rate is:

$$\dot{E_x} = \dot{m}ex \tag{173}$$

These relations are consistent with the exergy analysis formulation in Paper 2[1], where the specific flow exergy and exergy flow rate are defined using enthalpy and entropy relative to the dead state.

### 4-4-3 EES Simulation results

Parametric Table

Table 1

| | $T_{evp}$ | $Pevp_{D2}$ | $etaTh_{D2}$ | $etaEx_{D2}$ | $Wnet_{D2}$ | $mwf_{D2}$ | $Qevap_{D2}$ | $Tgeo_{out,D2}$ |
|---|---|---|---|---|---|---|---|---|
| Run 1 | 120 | 1201 | 13.78 | 38.17 | 49.54 | 1.799 | 359.6 | 96.11 |
| Run 2 | 125 | 1326 | 14.31 | 37.24 | 48.34 | 1.674 | 337.7 | 101.3 |
| Run 3 | 130 | 1461 | 14.82 | 35.86 | 46.54 | 1.544 | 314 | 106.9 |
| Run 4 | 135 | 1605 | 15.3 | 33.98 | 44.11 | 1.407 | 288.3 | 113 |
| Run 5 | 140 | 1760 | 15.75 | 31.58 | 40.98 | 1.261 | 260.3 | 119.6 |

*Figure 11: EES Simulation results in table format because of iterative solution*

### 4-4-4 Parametric Results

*Table 21: The EES results obtained by varying the evaporator temperature for part D_2*

| $T_{evp}$ $(°C)$ | $P_{evp}$ $(kPa)$ | $\eta_{th}$ (%) | $\eta_{ex}$ (%) | $\dot{W}_{net}$ $(kW)$ | $\dot{m}_{wf}$ $(kg/s)$ | $\dot{Q}_{evap}$ $(kW)$ | $T_{geo,out}$ $(°C)$ |
|---|---|---|---|---|---|---|---|
| 120 | 1201 | 13.78 | 38.17 | 49.54 | 1.799 | 359.6 | 96.11 |
| 125 | 1326 | 14.31 | 37.24 | 48.34 | 1.674 | 337.7 | 101.3 |
| 130 | 1461 | 14.82 | 35.86 | 46.54 | 1.544 | 314.0 | 106.9 |
| 135 | 1605 | 15.30 | 33.98 | 44.11 | 1.407 | 288.3 | 113.0 |
| 140 | 1760 | 15.75 | 31.58 | 40.98 | 1.261 | 260.3 | 119.6 |

## 4-4-5 Effect on Thermal Efficiency

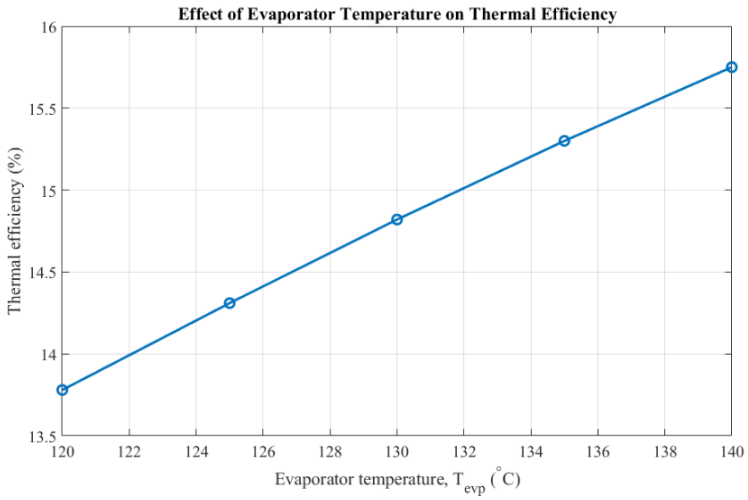



*Figure 12: Effect of evaporator temperature on thermal efficiency*

The thermal efficiency increases continuously as the evaporator temperature increases.

- At $T_{evp} = 120°C$ the thermal efficiency is $\eta_{th} = 13.78\%$
- At $T_{evp} = 140°C$ the thermal efficiency increases to $\eta_{th} = 15.75\%$

Therefore, the increase in thermal efficiency is $15.75 - 13.78 = 1.97$ percentage points.

The relative increase is:

$$\frac{15.75 - 13.78}{13.78} \times 100 = 14.3\% \tag{174}$$

This increase occurs because raising the evaporator temperature increases the average temperature of heat addition. From a first-law point of view, a higher heat addition temperature improves the ratio of net power output to heat input.

However, it is important to note that the net power output does not increase in this case. Instead, it decreases from 49.54 kW to 40.98 kW.

At the same time, the evaporator heat input decreases more strongly, from 359.6 kW to 260.3 kW.

The decrease in heat input is approximately:

$$\frac{359.6 - 260.3}{359.6} \times 100 = 27.6\% \tag{175}$$

while the decrease in net power output is approximately:

$$\frac{49.54 - 40.98}{49.54} \times 100 = 17.3\% \tag{176}$$

Since $\dot{Q_{ORC}}$ decreases more strongly than $\dot{W_{net}}$, the ratio $\dot{W_{net}}/\dot{Q_{ORC}}$ increases. Therefore, the thermal efficiency increases even though the absolute net power output decreases.

### 4-4-6 Effect on Exergy Efficiency

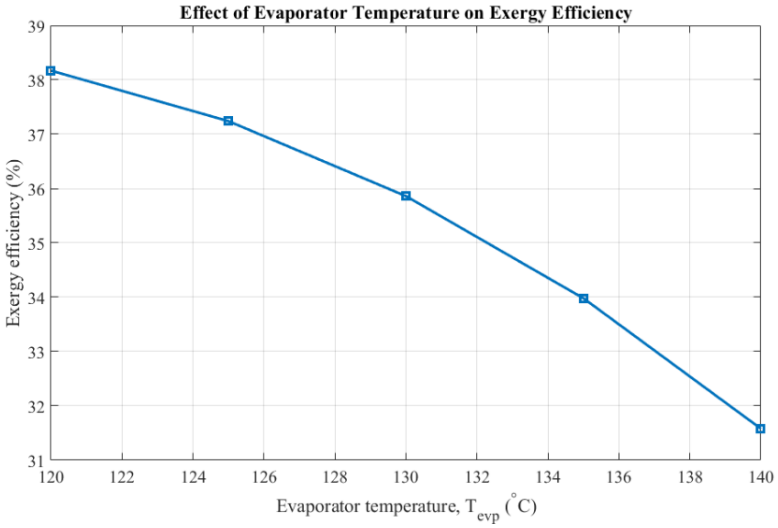



*Figure 13: Effect of evaporator temperature on exergy efficiency*

The exergy efficiency decreases as the evaporator temperature increases.

- At $T_{evp} = 120°C$ the exergy efficiency is $\eta_{ex} = 38.17\%$.
- At $T_{evp} = 140°C$ the exergy efficiency decreases to $\eta_{ex} = 31.58\%$.

The reduction in exergy efficiency is $38.17 - 31.58 = 6.59$ percentage points.

The reason for this decrease is related to the fixed geothermal inlet condition and the fixed pinch point temperature difference. In this parametric study $T_{geo} = 180°C$ and $\Delta T_{pinch} = 10°C$ are kept constant. When $T_{evp}$ increases, the pinch point temperature also increases:

$$T_{pp} = T_{evp} + \Delta T_{pinch} \tag{177}$$

Therefore, increasing $T_{evp}$ from $120°C$ to $140°C$ increases $T_{pp}$ from $130°C$ to $150°C$.

This limits the amount of heat that can be extracted from the geothermal fluid. As a result, the geothermal outlet temperature increases from $96.11°C$ to $119.6°C$.

This means that more useful thermal availability remains in the geofluid outlet stream and is not converted into power.

The working fluid mass flow rate also decreases significantly $\dot{m_{wf}} = 1.799$ kg/s at 120°C, while $\dot{m_{wf}} = 1.261$ kg/s at 140°C. The reduction is:

$$\frac{1.799 - 1.261}{1.799} \times 100 = 29.9\% \quad (178)$$

Since the geofluid exergy input is nearly constant but the net power output decreases, the exergy efficiency also decreases.

### 4-4-7 Comparison Between Thermal and Exergy Efficiency Trends

The results show two opposite trends $T_{evp} \uparrow \Rightarrow \eta_{th} \uparrow$ but $T_{evp} \uparrow \Rightarrow \eta_{ex} \downarrow$.

This behavior is not contradictory. Thermal efficiency only compares the net power output with the heat absorbed by the ORC working fluid:

$$\eta_{th} = \frac{\dot{W_{net}}}{\dot{Q_{ORC}}} \times 100 \quad (179)$$

Therefore, if $\dot{Q_{ORC}}$ decreases more rapidly than $\dot{W_{net}}$, the thermal efficiency increases.

Exergy efficiency, however, compares the net power output with the available exergy supplied by the geothermal source:

$$\eta_{ex} = \frac{\dot{W_{net}}}{\dot{E_{x.af}}} \times 100 \quad (180)$$

Because $\dot{E_{x,gf}}$ remains approximately fixed and $\dot{W_{net}}$ decreases, the exergy efficiency decreases.

This distinction is important. A higher thermal efficiency does not always mean that the system uses the geothermal resource more effectively. In this case, increasing $T_{evp}$ improves the thermal efficiency ratio, but it also leaves more heat unused in the geothermal outlet stream and reduces the net power output.

### 4-4-8 Discussion

The increase in evaporator temperature increases the evaporator pressure from 1201 kPa to 1760 kPa. This raises the turbine inlet pressure and improves the thermodynamic quality of the heat addition process. Therefore, the thermal efficiency increases.

However, because the geothermal inlet temperature is fixed at $180°C$, increasing $T_{evp}$ reduces the available temperature difference between the geothermal fluid and the working fluid in the evaporator. Under the fixed pinch point constraint, this causes a reduction in heat recovery from the geothermal fluid.

The EES results confirm this behavior. The evaporator heat duty decreases from 359.6kW to 260.3kW , and the working fluid mass flow rate decreases from 1.799 kg/s to 1.261 kg/s. Consequently, the net power output decreases from 49.54 kW to 40.98 kW .

Therefore, increasing the evaporator temperature improves thermal efficiency but reduces the overall exergetic utilization of the geothermal source. This result shows that the evaporator

temperature should not be selected only based on thermal efficiency. A complete design decision must consider thermal efficiency, exergy efficiency, net power output, heat recovery, and geothermal outlet temperature together.

Paper 2[1] also emphasizes that evaporator pressure, condenser temperature, superheating degree, and pinch point temperature difference are important optimization variables in ORC and IHE-ORC systems. The present result supports this idea because changing only the evaporator temperature produces opposite trends in first-law and second-law performance.

### 4-4-9 Conclusion

In this section, the evaporator temperature was varied from $120°C$ to $140°C$. The results show that thermal efficiency increases, while exergy efficiency decreases.

The thermal efficiency increases from $\%13.78\%$ to $15.75\%$ because the evaporator heat input decreases more strongly than the net power output.

The exergy efficiency decreases from $38.17\%$ to $31.58\%$ because the net power output decreases while the geofluid exergy input remains nearly constant.

- The working fluid mass flow rate decreases from $1.799\ \mathrm{kg/s}$ to $1.261\ \mathrm{kg/s}$.
- and the net power output decreases from $49.54\ \mathrm{kW}$ to $49.98\ \mathrm{kW}$.

Therefore, although increasing the evaporator temperature improves the thermal efficiency, it weakens the exergy efficiency and reduces power generation. The best evaporator temperature should be selected by balancing both thermal and exergetic performance, not by considering thermal efficiency alone.

## 4-5 D-3) Effect of Condenser Temperature on Thermal Efficiency, Exergy Efficiency, and Total Product Cost Rate

### 4-5-1 Objective

The objective of this section is to investigate the effect of condenser temperature on the thermodynamic and economic performance of the ORC with Internal Heat Exchanger. In this parametric study, the condenser temperature was varied from $30°C \leq T_{cond} \leq 50°C$ while the evaporator temperature, geothermal inlet temperature, pinch point temperature difference, turbine efficiency, pump efficiency, IHE effectiveness, and geothermal mass flow rate were kept constant.

The investigated output parameters are $\eta_{th}\eta_{ex}\dot{C_{pro}}$ where $\eta_{th}$ is the thermal efficiency, $\eta_{ex}$ is the exergy efficiency, and $\dot{C_{pro}}$ is the total product cost rate.

### 4-5-2 Governing Equations

The condenser pressure is obtained from the saturation pressure of R123 at the condenser temperature:

$$P_{cond} = P_{sat,R123}(T_{cond}) \tag{181}$$

Therefore, increasing $T_{cond}$increases the condenser pressure. In the EES results, the condenser pressure increases from $P_{cond} = 109.7\ kPa$ at $T_{cond} = 30°C$, to $P_{cond} = 212.7\ kPa$ at $T_{cond} = 50°C$.

The net power output is calculated as:

$$\dot{W_{net}} = \dot{W}_T - \dot{W}_P \tag{182}$$

The thermal efficiency is:

$$\eta_{th} = \frac{\dot{W_{net}}}{\dot{Q_{ORC}}} \times 100 \tag{183}$$

The exergy efficiency is:

$$\eta_{ex} = \frac{\dot{W_{net}}}{\dot{E_{x,af}}} \times 100 \tag{184}$$

The total product cost rate is calculated from:

$$\dot{C_{pro}} = \dot{C_{fuel}} + \dot{Z_{tot}} \tag{185}$$

where $\dot{C_{fuel}}$ is the fuel cost rate associated with the geofluid exergy input and $\dot{Z_{tot}}$ is the total annualized equipment cost rate.

## 4-5-3 EES Simulation results

Table 1

| 1..5 | $T_{cond}$ | $Pcond_{D3}$ | $etaTh_{D3}$ | $etaEx_{D3}$ | $CdotPro_{D3}$ | $CproGJ_{D3}$ | $Wnet_{D3}$ | $Qcond_{D3}$ | $Acnd_{D3}$ | $mwf_{D3}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| Run 1 | 30 | 109.7 | 15.35 | 44.57 | 0.002493 | 43.09 | 57.85 | 319 | 240 | 1.799 |
| Run 2 | 35 | 130.7 | 14.57 | 41.32 | 0.001912 | 35.65 | 53.63 | 314.6 | 125 | 1.799 |
| Run 3 | 40 | 154.7 | 13.78 | 38.17 | 0.001651 | 33.33 | 49.54 | 310.1 | 83.81 | 1.799 |
| Run 4 | 45 | 181.9 | 12.98 | 35.11 | 0.001482 | 32.52 | 45.58 | 305.4 | 62.6 | 1.799 |
| Run 5 | 50 | 212.7 | 12.19 | 32.15 | 0.001353 | 32.42 | 41.73 | 300.7 | 49.65 | 1.799 |

*Figure 14: EES Simulation results for section D_3 of project*

## 4-5-4 Parametric Results

*Table 22: EES results for different condenser temperatures for section D_3 of project*

| $T_{cond}T$ $(°C)$ | $P_{cond}$ $(kPa)$ | $\eta_{th}$ (%) | $\eta_{ex}$ (%) | $\dot{C_{pro}}$ $(\$/s)$ | $\dot{C_{pro}}$ $(\$/h)$ | $c_{pro}$ $(\$/GJ)$ | $\dot{W_{net}}t$ $(kW)$ | $Q_{cond}$ $(kW)$ | $A_{cond}$ $(m^2)$ |
|---|---|---|---|---|---|---|---|---|---|
| 30 | 109.7 | 15.35 | 44.57 | 0.002493 | 8.975 | 43.09 | 57.85 | 319.0 | 240.0 |
| 35 | 130.7 | 14.57 | 41.32 | 0.001912 | 6.883 | 35.65 | 53.63 | 314.6 | 125.0 |
| 40 | 154.7 | 13.78 | 38.17 | 0.001651 | 5.944 | 33.33 | 49.54 | 310.1 | 83.81 |
| 45 | 181.9 | 12.98 | 35.11 | 0.001482 | 5.335 | 32.52 | 45.58 | 305.4 | 62.60 |
| 50 | 212.7 | 12.19 | 32.15 | 0.001353 | 4.871 | 32.42 | 41.73 | 300.7 | 49.65 |

## 4-5-5 Effect on Thermal Efficiency

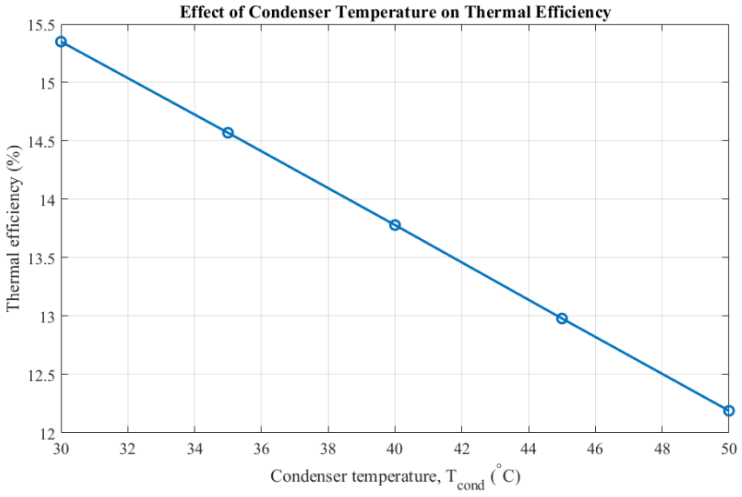



*Figure 15: Effect of condenser temperature on thermal efficiency*

The thermal efficiency decreases continuously as the condenser temperature increases.

- At $T_{cond} = 30°C$ the thermal efficiency is $\eta_{th} = 15.35\%$.
- At $T_{cond} = 50°C$ the thermal efficiency decreases to $\eta_{th} = 12.19\%$.

The reduction is $15.35 - 12.19 = 3.16$ percentage points.

This decrease is physically expected. Increasing the condenser temperature increases the condenser pressure. As a result, the turbine outlet pressure increases and the turbine pressure ratio decreases. Therefore, the turbine produces less work. The net power output decreases from 57.85 kW to 41.73 kW.

Since thermal efficiency depends directly on net power output, the reduction in $\dot{W_{net}}$net causes the thermal efficiency to decrease.

## 4-6 Effect on Exergy Efficiency

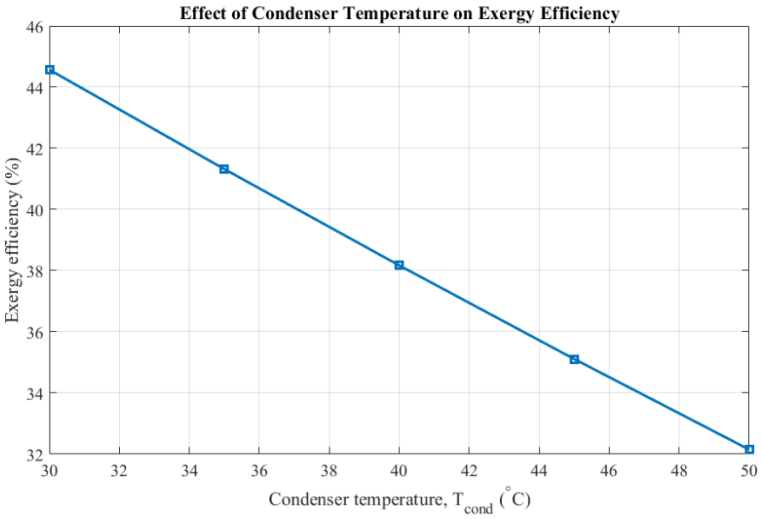



*Figure 16: Effect of condenser temperature on exergy efficiency*

The exergy efficiency also decreases as the condenser temperature increases.

- At $T_{cond} = 30°C$ the exergy efficiency is $\eta_{ex} = 44.57\%$
- At $T_{cond} = 50°C$ it decreases to $\eta_{ex} = 32.15\%$

The reduction in exergy efficiency is $44.57 - 32.15 = 12.42$ percentage points.

This behavior occurs because the geofluid inlet condition is fixed in this parametric study. Therefore, the geothermal exergy input remains almost constant. However, the useful output of the system, represented by $\dot{W_{net}}$, decreases significantly as the condenser temperature increases. Since:

$$\eta_{ex} = \frac{\dot{W_{net}}}{\dot{E_{x,gf}}} \times 100 \tag{186}$$

a reduction in $\dot{W_{net}}$ directly reduces the exergy efficiency.

### 4-6-1 Effect on Total Product Cost Rate

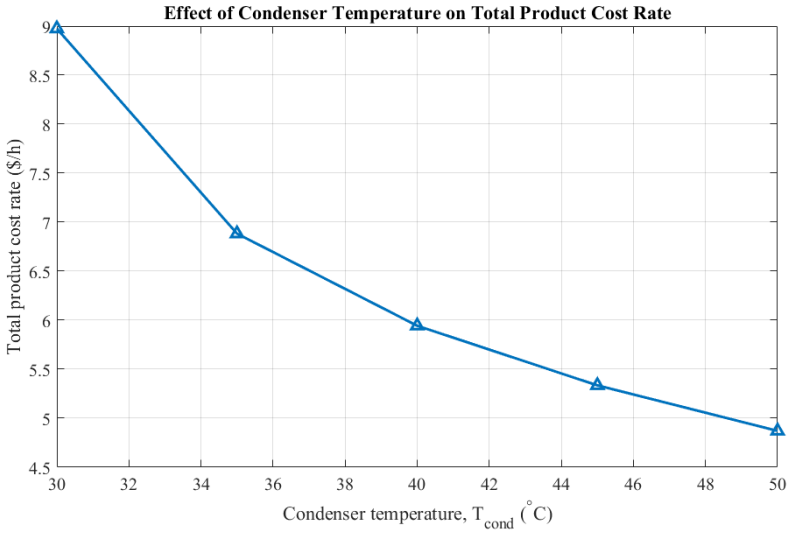



*Figure 17: Effect of condenser temperature on the total product cost rate*

The total product cost rate decreases as the condenser temperature increases.

- At $T_{cond} = 30°C$ the total product cost rate is $\dot{C}_{pro} = 0.002493$ \$/s or 8.975 \$/$h$.
- At $T_{cond} = 50°C$ it decreases to $\dot{C}_{pro} = 0.001353$ \$/$s$ or 4.871 \$/$h$.

The decrease in total product cost rate is mainly caused by the reduction in condenser area. As the condenser temperature increases, the terminal temperature differences in the condenser become larger. Therefore, the required condenser area decreases from $A_{cond} = 240.0$ m$^2$ to $A_{cond} = 49.65$ m$^2$.

This significant reduction in condenser area reduces the condenser capital cost and consequently reduces the total annualized equipment cost rate. Therefore, even though the cycle produces less power at higher condenser temperatures, the total cost rate per unit time decreases.

### 4-6-2 Discussion

The results show that increasing the condenser temperature has a negative effect on the thermodynamic performance of the ORC-IHE system. Both thermal efficiency and exergy efficiency decrease as $T_{cond}$ increases.

This happens because the condenser temperature determines the low-pressure level of the ORC. A higher condenser temperature causes a higher condenser pressure. This reduces the turbine expansion ratio and decreases the turbine work output. Since the working fluid mass flow rate remains constant at $\dot{m}_{wf} = 1.799$ kg/s the reduction in turbine specific work directly reduces the total net power output.

The constant working fluid mass flow rate is reasonable in this parametric study because $\dot{m_{wf}}$ is determined from the evaporator pinch-point energy balance. Since $T_{evp}$, $T_{geo}$, $\Delta T_{pinch}$, and $\dot{m_{geo}}$ remain constant, the working fluid mass flow rate does not change.

From the economic viewpoint, the trend is different. The total product cost rate decreases with increasing condenser temperature. This does not mean that the thermodynamic performance improves. Instead, it means that the required condenser area becomes much smaller, which reduces equipment cost. The condenser area decreases by approximately:

$$\frac{240.0 - 49.65}{240.0} \times 100 = 79.3\% \tag{187}$$

This reduction strongly affects the economic results because the condenser is one of the major heat transfer components in the system.

However, it is important to distinguish between total product cost rate and specific production cost. The total product cost rate, $\dot{C_{pro}}$, represents cost per unit time. The specific production cost, $c_{pro}$, represents cost per unit produced energy. In this simulation, both decrease as $T_{cond}$ increases, but the thermodynamic efficiencies decrease. Therefore, the economic and thermodynamic trends do not lead to the same design conclusion.

### 4-6-3 Conclusion

In this section, the condenser temperature was varied from $30°C$ to $50°C$. The results show that increasing condenser temperature decreases the thermal efficiency, exergy efficiency, net power output, condenser area, and total product cost rate.

- The thermal efficiency decreases from $15.35\%$ to$12.19\%$.
- The exergy efficiency decreases from $44.57\%$ to $32.15\%$
- The net power output decreases from $57.85$ kW to $41.73$ kW.
- The total product cost rate decreases from $0.002493\$/s$ to $0.001353\$/s$ or, in hourly form, from $8.975\ \$/h$ to $4.871\ \$/h$.

Therefore, a lower condenser temperature is better from the thermodynamic viewpoint because it increases net power output and improves both thermal and exergy efficiencies. However, a higher condenser temperature reduces the required condenser area and lowers the total product cost rate. This shows a trade-off between thermodynamic performance and equipment cost. The final selection of condenser temperature should therefore be based on both efficiency and economic criteria

# Part E: Validation

## 5-1 Objective

The objective of this section is to validate the EES simulation results by comparing them with the data and findings reported in Paper 1[2]. The validation is performed for the ORC with Internal Heat Exchanger using R123 as the working fluid.

The comparison focuses on four main aspects $\dot{W_{net}}, \eta_{I,1}, \eta_{I,2}, \dot{E_{D,total}}$.In addition, the exergy destruction distribution among the system components is compared with the exergy destruction ratios reported in Paper 1.

Paper 1 uses the same main modeling assumptions used in this project: steady-state operation, negligible pressure drops in heat exchangers and pipelines, isentropic efficiencies for turbines and pumps, negligible kinetic and potential energy changes, saturated-liquid geofluid at the reservoir, and water properties for the geofluid.

## 5-2 Validation Basis

The input conditions used in this project are consistent with Paper 1[2]. The main operating conditions are:

*Table 23: The main operating conditions*

| Parameter | Value |
|---|---|
| Dead − state pressure, $\mathbf{P_0}$ | 100 kPa |
| Dead − state temperature, $\mathbf{T_0}$ | 25°C |
| Condenser temperature, $\mathbf{T_C}$ | 40°C |
| Evaporator temperature, $\mathbf{T_E}$ | 120°C |
| Geo fluid inlet temperature, $\mathbf{T_{geo}}$ | 180°C |
| Pump efficiency, $\boldsymbol{\eta_P}$ | 90% |
| Turbine efficiency, $\boldsymbol{\eta_T}$ | 80% |
| Pinch point temperature difference, $\boldsymbol{\Delta T_E}$ | 10 K |

These values match the simulation parameters reported in Table 2 of Paper 1[2].

Paper 1[2] defines the specific flow exergy as:

$$e = (h - h_0) - T_0(s - s_0) \tag{188}$$

and the exergy rate as:

$$\dot{E} = \dot{m}e\dot{E} \tag{189}$$

The first-law and second-law efficiencies are then defined based on the ratio of product energy or exergy to the corresponding input energy or exergy. These are the same definitions used in the present EES model.

## 5-3 State Property Validation

Because the state numbering in the present model is not exactly identical to all tables in Paper 1[2], the validation was first performed using equivalent boundary states and operating conditions.

*Table 24: The key thermodynamic properties*

| Quantity | Present EES result | Paper 1 value | Comment |
|---|---|---|---|
| $P_{cond}$ | 154.7 kPa | 154.7 kPa | Matched |
| $P_{evp}$ | 1201 kPa | 1201 kPa | Matched |
| h of saturated liquid R123 at | 241.9 kJ/kg | 241.9 kJ/kg | Matched |
| s of saturated liquid R123 at 40°C | 1.142 kJ/kg. K | 1.143 kJ/kg. K | Very close |
| h of saturated vapor R123 at 120°C | 452.9 kJ/kg | 452.9 kJ/kg | Matched |
| s of saturated vapor R123 at 120°C | 1.702 kJ/kg. K | 1.702 kJ/kg. K | Matched |
| Geofluid inlet enthalpy | 763.1 kJ/kg | 763.2 kJ/kg | Very close |
| Geofluid inlet entropy | 2.139 kJ/kg. K | 2.14 kJ/kg. K | Very close |

The small differences are due to rounding and possible differences in the thermodynamic property database. Overall, the state-property comparison confirms that the implemented EES model is consistent with the thermodynamic data used in Paper 1[2].

## 5-4 Performance Validation

Paper 1[2] reports the performance of the ORC with IHE for different working fluids in Table 7. For R123, the reported values are used as the reference for validation. In Paper 1[2], the net work and total exergy destruction are reported per unit geofluid mass flow rate. Since the present simulation uses $\dot{m_{geo}} = 1\ \mathrm{kg/s}$ the numerical values in kW are directly comparable with the values in kJ/kg.

The relative error is calculated as:

$$\text{Error} = \frac{\text{Present result} - \text{Paper 1result}}{\text{Paper 1 result}} \times 100 \tag{190}$$

*Table 25: Performance validation of the present ORC-IHE model against Paper 1 for R123*

| *Parameter* | *Present result* | *Paper 1 result* | *Relative error* |
|---|---|---|---|
| $\dot{W_{net}}$ | 49.54 $kW$ | 50.38 $kJ/kg$ | $-1.67\%$ |
| $\dot{E_{D,total}}$ | 80.25 $kW$ | 80.25 $kJ/kg$ | 0.00% |
| $\eta_{I,1}$ | 7.528% | 7.65% | $-1.59\%$ |
| $\eta_{I,2}$ | 13.78% | 14.20% | $-2.96\%$ |
| $\eta_{II,1}$ | 38.17% | 38.76% | $-1.52\%$ |
| $\eta_{II,2}$ | 50.47% | 51.40% | $-1.81\%$ |

Paper 1[2] reports for the ORC with IHE using R123: $\dot{W_{net}} = 50.38, \dot{E_D} = 80.25\%, \eta_{I,1} = 7.65\%, \eta_{I,2} = 14.20\%\eta, \eta_{II,1} = 38.76\%, \eta_{II,2} = 51.40\%$.

The agreement is very good. The largest relative deviation is less than 3%, which is acceptable for an independently reproduced thermodynamic model. The exact match in total exergy destruction confirms that the exergy balance was implemented correctly.

## 5-5 Exergy Destruction Distribution Validation

The component-level exergy destruction distribution was also compared with Fig. 5 of Paper 1[2]. In Paper 1[2], the exergy destruction of each component is reported as a percentage of the geofluid exergy input. The same ratio was calculated here as:

$$Y_{D,i} = \frac{\dot{E}_{D,i}}{\dot{E}_{x,gf}} \times 100 \tag{191}$$

Using $\dot{E}_{x,gf} = 129.8$ kW the present exergy destruction ratios are:

*Table 26: Exergy Destruction Distribution Validation*

| Component | Present result YDY_DYD (%) | Paper 1 result YDY_DYD (%) | Difference |
|---|---|---|---|
| Condenser | 8.05 | 7.797 | +0.25 |
| Evaporator | 16.46 | 16.52 | −0.06 |
| IHE | 0.48 | 0.42 | +0.06 |
| Pump | 0.11 | 0.10 | +0.01 |
| Reinjection | 24.38 | 24.59 | −0.21 |
| Turbine | 8.78 | 8.799 | −0.02 |
| Cooling air | 3.59 | 3.512 | +0.07 |
| **Total** | **61.83** | **61.74** | **+0.09** |

The component-level agreement is very strong. Paper 1[2] shows that for the ORC with IHE using R123, the largest exergy destruction is associated with reinjection, followed by the evaporator, turbine, condenser, cooling air, IHE, and pump. The present model predicts the same order of dominant irreversibilities.

## 5-6 Discussion

The validation results show that the present EES model reproduces the main findings of Paper 1[2] with good accuracy. The net power output differs from the reference value by only 1.67%, while the total exergy destruction rate matches the reference value exactly. The first-law and second-law efficiencies also show small deviations, all below 3%.

The small differences may be caused by several factors. First, EES property data may slightly differ depending on the database version. Second, the values reported in Paper 1[2] are rounded in tables and figures. Third, the present model follows the ORC-IHE configuration directly, while some state-property tables in Paper 1[2] include regenerative cycle states, so only equivalent states and performance parameters should be compared.

The exergy destruction distribution confirms the physical correctness of the model. In both the present simulation and Paper 1[2], the reinjection stream is the largest source of exergy loss. This means that a significant part of the geothermal exergy remains unused in the geofluid leaving the evaporator. The evaporator is the second major source of exergy destruction because heat transfer occurs across a finite temperature difference between the geofluid and the working fluid. The

turbine and condenser also have significant irreversibilities, while the pump and IHE have relatively small contributions.

This agreement validates not only the numerical results but also the thermodynamic behavior predicted by the model.

## 5-7 Conclusion

The EES model developed in this project was validated against Paper 1[2] for the ORC with IHE using R123 as the working fluid. The comparison shows that the present results are in close agreement with the reference data.

- The main validation results are $\dot{W}_{net} = 49.54$ kW compared with $50.38$ kJ/kg from Paper 1[2].
- The total exergy destruction rate is $\dot{E}_{D,total} = 80.25$ kW which exactly matches the value reported in Paper 1[2] for $\dot{m}_{geo} = 1$ kg/s.
- The exergy efficiency based on geofluid exergy input is $\eta_{II,1} = 38.17\%$compared with 38.76% from Paper1.
- The component exergy destruction distribution also matches the trend reported in Paper 1[2].
- Reinjection has the highest exergy destruction, followed by the evaporator, turbine, condenser, cooling air, IHE, and pump.

Therefore, the developed EES model is considered validated and reliable for the subsequent heat exchanger area calculation, economic analysis, and parametric studies.

# References

[1] S. Mohammadzadeh Bina, S. Jalilinasrabady, and H. Fujii, "Energy, economic and environmental (3E) aspects of internal heat exchanger for ORC geothermal power plants," *Energy*, vol. 140, pp. 1096–1106, 2017, doi: 10.1016/j.energy.2017.09.045.

[2] M. Yari, "Exergetic analysis of various types of geothermal power plants," *Renew. Energy*, vol. 35, no. 1, pp. 112–121, Jan. 2010, doi: 10.1016/j.renene.2009.07.023.

# Appendix A: EES code for section A_1

```
{Input data}

P0 = 100
T0 = 25

T_cond = 40
T_evp = 120
T_geo = 180

DT_pinch = 10

eta_T = 0.80
eta_P = 0.90

eps_IHE = 0.60

m_geo = 1


{--------------------}
{Dead state}
{--------------------}

h0 = enthalpy(Water,T=T0,P=P0)
s0 = entropy(Water,T=T0,P=P0)


{--------------------}
{Cycle pressures}
{--------------------}

P[1] = pressure(R123,T=T_cond,x=0)
P[2] = pressure(R123,T=T_evp,x=0)

P[3] = P[2]
P[4] = P[2]

P[5] = P[1]
P[6] = P[1]

P[9] = P0
P[10] = P0


{--------------------}
{State 1: condenser outlet / pump inlet}
{--------------------}

T[1] = T_cond
x[1] = 0

h[1] = enthalpy(R123,T=T[1],x=x[1])
s[1] = entropy(R123,T=T[1],x=x[1])
v[1] = volume(R123,T=T[1],x=x[1])


{--------------------}
{State 2: pump outlet / IHE cold-side inlet}
{--------------------}

h_2s = h[1] + v[1]*(P[2]-P[1])
h[2] = h[1] + (h_2s-h[1])/eta_P

T[2] = temperature(R123,P=P[2],h=h[2])
s[2] = entropy(R123,P=P[2],h=h[2])


{--------------------}
{State 4: evaporator outlet / turbine inlet}
{--------------------}
```

```
T[4] = T_evp
x[4] = 1

h[4] = enthalpy(R123,T=T[4],x=x[4])
s[4] = entropy(R123,T=T[4],x=x[4])


{--------------------}
{State 5: turbine outlet / IHE hot-side inlet}
{--------------------}

s_5s = s[4]
h_5s = enthalpy(R123,P=P[5],s=s_5s)

h[5] = h[4] - eta_T*(h[4]-h_5s)

T[5] = temperature(R123,P=P[5],h=h[5])
s[5] = entropy(R123,P=P[5],h=h[5])


{--------------------}
{State 6: IHE hot-side outlet / condenser inlet}
{--------------------}

T[6] = T[5] - eps_IHE*(T[5]-T[2])

h[6] = enthalpy(R123,P=P[6],T=T[6])
s[6] = entropy(R123,P=P[6],T=T[6])


{--------------------}
{State 3: IHE cold-side outlet / evaporator inlet}
{--------------------}

h[3] = h[2] + h[5] - h[6]

T[3] = temperature(R123,P=P[3],h=h[3])
s[3] = entropy(R123,P=P[3],h=h[3])


{--------------------}
{Geofluid side}
{--------------------}

T[7] = T_geo
x_geo_in = 0

P[7] = pressure(Water,T=T[7],x=x_geo_in)
P[8] = P[7]

h[7] = enthalpy(Water,T=T[7],x=x_geo_in)
s[7] = entropy(Water,T=T[7],x=x_geo_in)

T_pp = T_evp + DT_pinch
h_pp = enthalpy(Water,T=T_pp,P=P[7])

h_f_evp = enthalpy(R123,T=T_evp,x=0)

m_wf*(h[4]-h_f_evp) = m_geo*(h[7]-h_pp)

Q_evap = m_wf*(h[4]-h[3])

h[8] = h[7] - Q_evap/m_geo

T[8] = temperature(Water,P=P[8],h=h[8])
s[8] = entropy(Water,P=P[8],h=h[8])


{--------------------}
{Cooling air side}
{--------------------}

T[9] = T0
```

```
T[10] = T0 + 10

h[9] = enthalpy(Air,T=T[9])
h[10] = enthalpy(Air,T=T[10])

s[9] = entropy(Air,T=T[9],P=P[9])
s[10] = entropy(Air,T=T[10],P=P[10])

Q_cond = m_wf*(h[6]-h[1])

m_air = Q_cond/(h[10]-h[9])


{--------------------}
{Performance parameters}
{--------------------}

W_T = m_wf*(h[4]-h[5])
W_P = m_wf*(h[2]-h[1])
W_net = W_T - W_P

Q_IHE = m_wf*(h[5]-h[6])
Q_ORC = m_wf*(h[4]-h[3])

eta_ORC = W_net/Q_ORC


{--------------------}
{Output variables for property table}
{--------------------}

T_1 = T[1]
T_2 = T[2]
T_3 = T[3]
T_4 = T[4]
T_5 = T[5]
T_6 = T[6]
T_7 = T[7]
T_8 = T[8]
T_9 = T[9]
T_10 = T[10]

P_1 = P[1]
P_2 = P[2]
P_3 = P[3]
P_4 = P[4]
P_5 = P[5]
P_6 = P[6]
P_7 = P[7]
P_8 = P[8]
P_9 = P[9]
P_10 = P[10]

h_1 = h[1]
h_2 = h[2]
h_3 = h[3]
h_4 = h[4]
h_5 = h[5]
h_6 = h[6]
h_7 = h[7]
h_8 = h[8]
h_9 = h[9]
h_10 = h[10]

s_1 = s[1]
s_2 = s[2]
s_3 = s[3]
s_4 = s[4]
s_5 = s[5]
s_6 = s[6]
s_7 = s[7]
s_8 = s[8]
s_9 = s[9]
s_10 = s[10]
```

```
{--------------------}
{Reference values}
{--------------------}

P_cond = P[1]
P_evp = P[4]

T_dead = T0
P_dead = P0
h_dead = h0
s_dead = s0
```

# Appendix B: EES code for section A_2

```
{Input data}
{--------------------}

P0 = 100
T0 = 25

T_cond = 40
T_evp = 120
T_geo = 180

DT_pinch = 10

eta_T = 0.80
eta_P = 0.90

eps_IHE = 0.60

m_geo = 1


{--------------------}
{Dead state}
{--------------------}

h0 = enthalpy(Water,T=T0,P=P0)
s0 = entropy(Water,T=T0,P=P0)


{--------------------}
{Cycle pressures}
{--------------------}

P[1] = pressure(R123,T=T_cond,x=0)
P[2] = pressure(R123,T=T_evp,x=0)

P[3] = P[2]
P[4] = P[2]

P[5] = P[1]
P[6] = P[1]

P[9] = P0
P[10] = P0


{--------------------}
{State 1: condenser outlet / pump inlet}
{--------------------}

T[1] = T_cond
x[1] = 0

h[1] = enthalpy(R123,T=T[1],x=x[1])
s[1] = entropy(R123,T=T[1],x=x[1])
v[1] = volume(R123,T=T[1],x=x[1])


{--------------------}
{State 2: pump outlet / IHE cold-side inlet}
{--------------------}

h_2s = h[1] + v[1]*(P[2]-P[1])
h[2] = h[1] + (h_2s-h[1])/eta_P

T[2] = temperature(R123,P=P[2],h=h[2])
s[2] = entropy(R123,P=P[2],h=h[2])


{--------------------}
{State 4: evaporator outlet / turbine inlet}
{--------------------}
```

```
T[4] = T_evp
x[4] = 1

h[4] = enthalpy(R123,T=T[4],x=x[4])
s[4] = entropy(R123,T=T[4],x=x[4])


{--------------------}
{State 5: turbine outlet / IHE hot-side inlet}
{--------------------}

s_5s = s[4]
h_5s = enthalpy(R123,P=P[5],s=s_5s)

h[5] = h[4] - eta_T*(h[4]-h_5s)

T[5] = temperature(R123,P=P[5],h=h[5])
s[5] = entropy(R123,P=P[5],h=h[5])


{--------------------}
{State 6: IHE hot-side outlet / condenser inlet}
{--------------------}

T[6] = T[5] - eps_IHE*(T[5]-T[2])

h[6] = enthalpy(R123,P=P[6],T=T[6])
s[6] = entropy(R123,P=P[6],T=T[6])


{--------------------}
{State 3: IHE cold-side outlet / evaporator inlet}
{--------------------}

h[3] = h[2] + h[5] - h[6]

T[3] = temperature(R123,P=P[3],h=h[3])
s[3] = entropy(R123,P=P[3],h=h[3])


{--------------------}
{Geofluid side}
{--------------------}

T[7] = T_geo
x_geo_in = 0

P[7] = pressure(Water,T=T[7],x=x_geo_in)
P[8] = P[7]

h[7] = enthalpy(Water,T=T[7],x=x_geo_in)
s[7] = entropy(Water,T=T[7],x=x_geo_in)

T_pp = T_evp + DT_pinch
h_pp = enthalpy(Water,T=T_pp,P=P[7])

h_f_evp = enthalpy(R123,T=T_evp,x=0)

m_wf*(h[4]-h_f_evp) = m_geo*(h[7]-h_pp)

Q_evap = m_wf*(h[4]-h[3])

h[8] = h[7] - Q_evap/m_geo

T[8] = temperature(Water,P=P[8],h=h[8])
s[8] = entropy(Water,P=P[8],h=h[8])


{--------------------}
{Cooling air side}
{--------------------}

T[9] = T0
```

```
T[10] = T0 + 10

h[9] = enthalpy(Air,T=T[9])
h[10] = enthalpy(Air,T=T[10])

s[9] = entropy(Air,T=T[9],P=P[9])
s[10] = entropy(Air,T=T[10],P=P[10])

Q_cond = m_wf*(h[6]-h[1])

m_air = Q_cond/(h[10]-h[9])


{--------------------}
{Performance parameters}
{--------------------}

W_T = m_wf*(h[4]-h[5])
W_P = m_wf*(h[2]-h[1])
W_net = W_T - W_P

Q_IHE = m_wf*(h[5]-h[6])
Q_ORC = m_wf*(h[4]-h[3])

eta_ORC = W_net/Q_ORC


{--------------------}
{Output variables for property table}
{--------------------}

T_1 = T[1]
T_2 = T[2]
T_3 = T[3]
T_4 = T[4]
T_5 = T[5]
T_6 = T[6]
T_7 = T[7]
T_8 = T[8]
T_9 = T[9]
T_10 = T[10]

P_1 = P[1]
P_2 = P[2]
P_3 = P[3]
P_4 = P[4]
P_5 = P[5]
P_6 = P[6]
P_7 = P[7]
P_8 = P[8]
P_9 = P[9]
P_10 = P[10]

h_1 = h[1]
h_2 = h[2]
h_3 = h[3]
h_4 = h[4]
h_5 = h[5]
h_6 = h[6]
h_7 = h[7]
h_8 = h[8]
h_9 = h[9]
h_10 = h[10]

s_1 = s[1]
s_2 = s[2]
s_3 = s[3]
s_4 = s[4]
s_5 = s[5]
s_6 = s[6]
s_7 = s[7]
s_8 = s[8]
s_9 = s[9]
s_10 = s[10]
```

```
{--------------------}
{Reference values}
{--------------------}

P_cond = P[1]
P_evp = P[4]

T_dead = T0
P_dead = P0
h_dead = h0
s_dead = s0


{=================================================}
{Part A-2: Performance summary}
{=================================================}

T0K = T0 + 273.15


{--------------------}
{Specific flow exergy of geofluid}
{--------------------}

psi7 = (h[7]-h0) - T0K*(s[7]-s0)
psi8 = (h[8]-h0) - T0K*(s[8]-s0)

Ex7 = m_geo*psi7
Ex8 = m_geo*psi8


{--------------------}
{Specific flow exergy of cooling air}
{--------------------}

hAir0 = enthalpy(Air,T=T0)
sAir0 = entropy(Air,T=T0,P=P0)

psiA9 = (h[9]-hAir0) - T0K*(s[9]-sAir0)
psiA10 = (h[10]-hAir0) - T0K*(s[10]-sAir0)

ExA9 = m_air*psiA9
ExA10 = m_air*psiA10


{--------------------}
{Component exergy destruction rates}
{--------------------}

XDp = T0K*m_wf*(s[2]-s[1])

XDt = T0K*m_wf*(s[5]-s[4])

XDihe = T0K*m_wf*((s[3]-s[2]) + (s[6]-s[5]))

XDevp = T0K*(m_wf*(s[4]-s[3]) + m_geo*(s[8]-s[7]))

XDcond = T0K*(m_wf*(s[1]-s[6]) + m_air*(s[10]-s[9]))

XDrein = Ex8

XDair = ExA10


{--------------------}
{Total exergy destruction rate}
{--------------------}

XDsum = XDp + XDt + XDihe + XDevp + XDcond + XDrein + XDair

XDtot = Ex7 - W_net
```

```
{--------------------}
{Performance summary variables}
{--------------------}

etaTh = 100*W_net/Q_ORC

etaEx = 100*W_net/Ex7

mdot_wf = m_wf

Wnet_A2 = W_net

XDtot_A2 = XDtot


{--------------------}
{Additional useful values}
{--------------------}

etaPlant = 100*W_net/(m_geo*(h[7]-h0))

etaExDrop = 100*W_net/(Ex7-Ex8)
```

# Appendix C: MATLAB code for section A_3

```
clc
clear
close all
components = {'Condenser', 'Cooling Air', 'Evaporator', ...
              'IHE', 'Pump', 'Reinjection', 'Turbine'};
ED = [10.45, 4.655, 21.36, 0.6182, 0.1374, 31.64, 11.39];
ED_total = sum(ED);
ED_percent = 100 * ED / ED_total;
figure('Color','w','Position',[200 200 950 520])
bar(ED, 'LineWidth', 1.1)
grid on
box on
set(gca, 'XTick', 1:length(components), ...
         'XTickLabel', components, ...
         'FontName', 'Times New Roman', ...
         'FontSize', 12)
xtickangle(35)
ylabel('Exergy destruction rate, $\dot{E}_D$ (kW)', ...
       'FontName', 'Times New Roman', ...
       'FontSize', 13, ...
       'Interpreter', 'latex')
xlabel('Component', ...
       'FontName', 'Times New Roman', ...
       'FontSize', 13)
title('Exergy Destruction Rate of ORC-IHE Components', ...
      'FontName', 'Times New Roman', ...
      'FontSize', 14, ...
      'FontWeight', 'bold')
ylim([0 max(ED)*1.18])
for i = 1:length(ED)
    text(i, ED(i) + 0.8, sprintf('%.2f', ED(i)), ...
        'HorizontalAlignment', 'center', ...
        'FontName', 'Times New Roman', ...
        'FontSize', 11)
end
fprintf('Total exergy destruction rate = %.2f kW\n', ED_total)
result_table = table(components', ED', ED_percent', ...
    'VariableNames', {'Component','ED_kW','Share_percent'});
disp(result_table)
exportgraphics(gcf, 'A3_exergy_destruction_bar_chart.png', 'Resolution', 300)
```

# Appendix D: EES code for section C

```
{Input data}
{--------------------}

P0 = 100
T0 = 25

T_cond = 40
T_evp = 120
T_geo = 180

DT_pinch = 10

eta_T = 0.80
eta_P = 0.90

eps_IHE = 0.60

m_geo = 1


{--------------------}
{Dead state}
{--------------------}

h0 = enthalpy(Water,T=T0,P=P0)
s0 = entropy(Water,T=T0,P=P0)


{--------------------}
{Cycle pressures}
{--------------------}

P[1] = pressure(R123,T=T_cond,x=0)
P[2] = pressure(R123,T=T_evp,x=0)

P[3] = P[2]
P[4] = P[2]

P[5] = P[1]
P[6] = P[1]

P[9] = P0
P[10] = P0


{--------------------}
{State 1: condenser outlet / pump inlet}
{--------------------}

T[1] = T_cond
x[1] = 0

h[1] = enthalpy(R123,T=T[1],x=x[1])
s[1] = entropy(R123,T=T[1],x=x[1])
v[1] = volume(R123,T=T[1],x=x[1])


{--------------------}
{State 2: pump outlet / IHE cold-side inlet}
{--------------------}

h_2s = h[1] + v[1]*(P[2]-P[1])
h[2] = h[1] + (h_2s-h[1])/eta_P

T[2] = temperature(R123,P=P[2],h=h[2])
s[2] = entropy(R123,P=P[2],h=h[2])


{--------------------}
{State 4: evaporator outlet / turbine inlet}
{--------------------}
```

```
T[4] = T_evp
x[4] = 1

h[4] = enthalpy(R123,T=T[4],x=x[4])
s[4] = entropy(R123,T=T[4],x=x[4])


{--------------------}
{State 5: turbine outlet / IHE hot-side inlet}
{--------------------}

s_5s = s[4]
h_5s = enthalpy(R123,P=P[5],s=s_5s)

h[5] = h[4] - eta_T*(h[4]-h_5s)

T[5] = temperature(R123,P=P[5],h=h[5])
s[5] = entropy(R123,P=P[5],h=h[5])


{--------------------}
{State 6: IHE hot-side outlet / condenser inlet}
{--------------------}

T[6] = T[5] - eps_IHE*(T[5]-T[2])

h[6] = enthalpy(R123,P=P[6],T=T[6])
s[6] = entropy(R123,P=P[6],T=T[6])


{--------------------}
{State 3: IHE cold-side outlet / evaporator inlet}
{--------------------}

h[3] = h[2] + h[5] - h[6]

T[3] = temperature(R123,P=P[3],h=h[3])
s[3] = entropy(R123,P=P[3],h=h[3])


{--------------------}
{Geofluid side}
{--------------------}

T[7] = T_geo
x_geo_in = 0

P[7] = pressure(Water,T=T[7],x=x_geo_in)
P[8] = P[7]

h[7] = enthalpy(Water,T=T[7],x=x_geo_in)
s[7] = entropy(Water,T=T[7],x=x_geo_in)

T_pp = T_evp + DT_pinch
h_pp = enthalpy(Water,T=T_pp,P=P[7])

h_f_evp = enthalpy(R123,T=T_evp,x=0)

m_wf*(h[4]-h_f_evp) = m_geo*(h[7]-h_pp)

Q_evap = m_wf*(h[4]-h[3])

h[8] = h[7] - Q_evap/m_geo

T[8] = temperature(Water,P=P[8],h=h[8])
s[8] = entropy(Water,P=P[8],h=h[8])


{--------------------}
{Cooling air side}
{--------------------}

T[9] = T0
```

```
T[10] = T0 + 10

h[9] = enthalpy(Air,T=T[9])
h[10] = enthalpy(Air,T=T[10])

s[9] = entropy(Air,T=T[9],P=P[9])
s[10] = entropy(Air,T=T[10],P=P[10])

Q_cond = m_wf*(h[6]-h[1])

m_air = Q_cond/(h[10]-h[9])


{--------------------}
{Performance parameters}
{--------------------}

W_T = m_wf*(h[4]-h[5])
W_P = m_wf*(h[2]-h[1])
W_net = W_T - W_P

Q_IHE = m_wf*(h[5]-h[6])
Q_ORC = m_wf*(h[4]-h[3])

eta_ORC = W_net/Q_ORC


{--------------------}
{Output variables for property table}
{--------------------}

T_1 = T[1]
T_2 = T[2]
T_3 = T[3]
T_4 = T[4]
T_5 = T[5]
T_6 = T[6]
T_7 = T[7]
T_8 = T[8]
T_9 = T[9]
T_10 = T[10]

P_1 = P[1]
P_2 = P[2]
P_3 = P[3]
P_4 = P[4]
P_5 = P[5]
P_6 = P[6]
P_7 = P[7]
P_8 = P[8]
P_9 = P[9]
P_10 = P[10]

h_1 = h[1]
h_2 = h[2]
h_3 = h[3]
h_4 = h[4]
h_5 = h[5]
h_6 = h[6]
h_7 = h[7]
h_8 = h[8]
h_9 = h[9]
h_10 = h[10]

s_1 = s[1]
s_2 = s[2]
s_3 = s[3]
s_4 = s[4]
s_5 = s[5]
s_6 = s[6]
s_7 = s[7]
s_8 = s[8]
s_9 = s[9]
s_10 = s[10]
```

```
{--------------------}
{Reference values}
{--------------------}

P_cond = P[1]
P_evp = P[4]

T_dead = T0
P_dead = P0
h_dead = h0
s_dead = s0


{=================================================}
{Part A-2: Performance summary}
{=================================================}

T0K = T0 + 273.15


{--------------------}
{Specific flow exergy of geofluid}
{--------------------}

psi7 = (h[7]-h0) - T0K*(s[7]-s0)
psi8 = (h[8]-h0) - T0K*(s[8]-s0)

Ex7 = m_geo*psi7
Ex8 = m_geo*psi8


{--------------------}
{Specific flow exergy of cooling air}
{--------------------}

hAir0 = enthalpy(Air,T=T0)
sAir0 = entropy(Air,T=T0,P=P0)

psiA9 = (h[9]-hAir0) - T0K*(s[9]-sAir0)
psiA10 = (h[10]-hAir0) - T0K*(s[10]-sAir0)

ExA9 = m_air*psiA9
ExA10 = m_air*psiA10


{--------------------}
{Component exergy destruction rates}
{--------------------}

XDp = T0K*m_wf*(s[2]-s[1])

XDt = T0K*m_wf*(s[5]-s[4])

XDihe = T0K*m_wf*((s[3]-s[2]) + (s[6]-s[5]))

XDevp = T0K*(m_wf*(s[4]-s[3]) + m_geo*(s[8]-s[7]))

XDcond = T0K*(m_wf*(s[1]-s[6]) + m_air*(s[10]-s[9]))

XDrein = Ex8

XDair = ExA10


{--------------------}
{Total exergy destruction rate}
{--------------------}

XDsum = XDp + XDt + XDihe + XDevp + XDcond + XDrein + XDair

XDtot = Ex7 - W_net
```

```
{--------------------}
{Performance summary variables}
{--------------------}

etaTh = 100*W_net/Q_ORC

etaEx = 100*W_net/Ex7

mdot_wf = m_wf

Wnet_A2 = W_net

XDtot_A2 = XDtot


{--------------------}
{Additional useful values}
{--------------------}

etaPlant = 100*W_net/(m_geo*(h[7]-h0))

etaExDrop = 100*W_net/(Ex7-Ex8)

{=================================================}
{Part C: Economic analysis}
{=================================================}

{--------------------}
{Heat exchanger areas}
{--------------------}

UcondC = 250
UevapC = 850
UiheC = 300

DTc1 = T[6] - T[10]
DTc2 = T[1] - T[9]
LMTDc = (DTc1 - DTc2)/ln(DTc1/DTc2)
Acnd = Q_cond*1000/(UcondC*LMTDc)

DTe1 = T[7] - T[4]
DTe2 = T[8] - T[3]
LMTDe = (DTe1 - DTe2)/ln(DTe1/DTe2)
Aevp = Q_evap*1000/(UevapC*LMTDe)

DTi1 = T[5] - T[3]
DTi2 = T[6] - T[2]
LMTDi = (DTi1 - DTi2)/ln(DTi1/DTi2)
Aihx = Q_IHE*1000/(UiheC*LMTDi)


{--------------------}
{Economic parameters}
{--------------------}

oph = 7000
cgf = 1.3
life = 20
intr = 0.14
mf = 1.06

CRF = intr*(1 + intr)^life/((1 + intr)^life - 1)


{--------------------}
{Geofluid exergy input}
{--------------------}

T0KE = T0 + 273.15

psiGF = (h[7] - h0) - T0KE*(s[7] - s0)
ExGF = m_geo*psiGF
```

```
{--------------------}
{Capital investment cost of components}
{--------------------}

Ztur = 4405*W_T^0.89
Zpmp = 1120*W_P^0.8
Zcnd = 1397*Acnd^0.89
Zevp = 1397*Aevp^0.89
Zihx = 2143*Aihx^0.514

Ztot = Ztur + Zpmp + Zcnd + Zevp + Zihx


{--------------------}
{Cost rates of components}
{--------------------}

ZdotTur = Ztur*mf*CRF/(oph*3600)
ZdotPmp = Zpmp*mf*CRF/(oph*3600)
ZdotCnd = Zcnd*mf*CRF/(oph*3600)
ZdotEvp = Zevp*mf*CRF/(oph*3600)
ZdotIhx = Zihx*mf*CRF/(oph*3600)

ZdotTot = ZdotTur + ZdotPmp + ZdotCnd + ZdotEvp + ZdotIhx


{--------------------}
{Fuel cost rate}
{--------------------}

Cfuel = cgf*ExGF/1000000


{--------------------}
{Total product cost rate and specific production cost}
{--------------------}

CdotPro = Cfuel + ZdotTot

CproGJ = CdotPro/W_net*1000000

CproKWh = CproGJ*0.0036

CdotPro_h = CdotPro*3600
```

# Appendix E: EES code for secion D_1

```
P0 = 100
T0 = 25

T_cond = 40
T_evp = 120
T_geo = 180

}DT_pinch is defined in the Parametric Table{
}DT_pinch = 10{

eta_T = 0.80
eta_P = 0.90

eps_IHE = 0.60

m_geo = 1


{--------------------}
}Dead state{
{--------------------}

h0 = enthalpy)Water,T=T0,P=P0(
s0 = entropy)Water,T=T0,P=P0(


{--------------------}
}Cycle pressures{
{--------------------}

P = [1]pressure)R123,T=T_cond,x(0=
P = [2]pressure)R123,T=T_evp,x(0=

P[3] = P[2]
P[4] = P[2]

P[5] = P[1]
P[6] = P[1]

P[9] = P0
P[10] = P0


{--------------------}
}State 1: condenser outlet / pump inlet{
{--------------------}

T[1] = T_cond
x[1] = 0

h = [1]enthalpy)R123,T=T,[1]x=x[1([
s = [1]entropy)R123,T=T,[1]x=x[1([
v = [1]volume)R123,T=T,[1]x=x[1([


{--------------------}
}State 2: pump outlet / IHE cold-side inlet{
{--------------------}

h_2s = h[1] + v[1]*(P[2]-P[1])
h[2] = h[1] + (h_2s-h[1])/eta_P

T = [2]temperature)R123,P=P,[2]h=h[2([
s = [2]entropy)R123,P=P,[2]h=h[2([


{--------------------}
}State 4: evaporator outlet / turbine inlet{
{--------------------}

T[4] = T_evp
```

```
x[4] = 1

h = [4]enthalpy)R123,T=T,[4]x=x[4([
s = [4]entropy)R123,T=T,[4]x=x[4([


{--------------------}
}State 5: turbine outlet / IHE hot-side inlet{
{--------------------}

s_5s = s[4]
h_5s = enthalpy)R123,P=P,[5]s=s_5s(

h[5] = h[4] - eta_T*(h[4]-h_5s)

T = [5]temperature)R123,P=P,[5]h=h[5([
s = [5]entropy)R123,P=P,[5]h=h[5([


{--------------------}
}State 6: IHE hot-side outlet / condenser inlet{
{--------------------}

T[6] = T[5] - eps_IHE*(T[5]-T[2])

h = [6]enthalpy)R123,P=P,[6]T=T[6([
s = [6]entropy)R123,P=P,[6]T=T[6([


{--------------------}
}State 3: IHE cold-side outlet / evaporator inlet{
{--------------------}

h[3] = h[2] + h[5] - h[6]

T = [3]temperature)R123,P=P,[3]h=h[3([
s = [3]entropy)R123,P=P,[3]h=h[3([


{--------------------}
}Geofluid side{
{--------------------}

T[7] = T_geo
x_geo_in = 0

P = [7]pressure)Water,T=T,[7]x=x_geo_in(
P[8] = P[7]

h = [7]enthalpy)Water,T=T,[7]x=x_geo_in(
s = [7]entropy)Water,T=T,[7]x=x_geo_in(

T_pp = T_evp + DT_pinch
h_pp = enthalpy)Water,T=T_pp,P=P[7([

h_f_evp = enthalpy)R123,T=T_evp,x(0=

m_wf*(h[4]-h_f_evp) = m_geo*(h[7]-h_pp)

Q_evap = m_wf*(h[4]-h[3])

h[8] = h[7] - Q_evap/m_geo

T = [8]temperature)Water,P=P,[8]h=h[8([
s = [8]entropy)Water,P=P,[8]h=h[8([


{--------------------}
}Cooling air side{
{--------------------}

T[9] = T0
T[10] = T0 + 10
```

```
h = [9]enthalpy)Air,T=T[9([
h = [10]enthalpy)Air,T=T[10([

s = [9]entropy)Air,T=T,[9]P=P[9([
s = [10]entropy)Air,T=T,[10]P=P[10([

Q_cond = m_wf*(h[6]-h[1])

m_air = Q_cond/(h[10]-h[9])


{--------------------}
}Performance parameters{
{--------------------}

W_T = m_wf*(h[4]-h[5])
W_P = m_wf*(h[2]-h[1])
W_net = W_T - W_P

Q_IHE = m_wf*(h[5]-h[6])
Q_ORC = m_wf*(h[4]-h[3])

eta_ORC = W_net/Q_ORC


{--------------------}
}State point output variables{
{--------------------}

T_1 = T[1]
T_2 = T[2]
T_3 = T[3]
T_4 = T[4]
T_5 = T[5]
T_6 = T[6]
T_7 = T[7]
T_8 = T[8]
T_9 = T[9]
T_10 = T[10]

P_1 = P[1]
P_2 = P[2]
P_3 = P[3]
P_4 = P[4]
P_5 = P[5]
P_6 = P[6]
P_7 = P[7]
P_8 = P[8]
P_9 = P[9]
P_10 = P[10]

h_1 = h[1]
h_2 = h[2]
h_3 = h[3]
h_4 = h[4]
h_5 = h[5]
h_6 = h[6]
h_7 = h[7]
h_8 = h[8]
h_9 = h[9]
h_10 = h[10]

s_1 = s[1]
s_2 = s[2]
s_3 = s[3]
s_4 = s[4]
s_5 = s[5]
s_6 = s[6]
s_7 = s[7]
s_8 = s[8]
s_9 = s[9]
s_10 = s[10]
```

```
{--------------------}
}Reference values{
{--------------------}

P_cond = P[1]
P_evp = P[4]

T_dead = T0
P_dead = P0
h_dead = h0
s_dead = s0


T0K = T0 + 273.15


{--------------------}
}Specific flow exergy of geofluid{
{--------------------}

psi7 = (h[7]-h0) - T0K*(s[7]-s0)
psi8 = (h[8]-h0) - T0K*(s[8]-s0)

Ex7 = m_geo*psi7
Ex8 = m_geo*psi8


{--------------------}
}Specific flow exergy of cooling air{
{--------------------}

hAir0 = enthalpy)Air,T=T0(
sAir0 = entropy)Air,T=T0,P=P0(

psiA9 = (h[9]-hAir0) - T0K*(s[9]-sAir0)
psiA10 = (h[10]-hAir0) - T0K*(s[10]-sAir0)

ExA9 = m_air*psiA9
ExA10 = m_air*psiA10


{--------------------}
}Component exergy destruction rates{
{--------------------}

XDp = T0K*m_wf*(s[2]-s[1])

XDt = T0K*m_wf*(s[5]-s[4])

XDihe = T0K*m_wf*((s[3]-s[2]) + (s[6]-s[5]))

XDevp = T0K*(m_wf*(s[4]-s[3]) + m_geo*(s[8]-s[7]))

XDcond = T0K*(m_wf*(s[1]-s[6]) + m_air*(s[10]-s[9]))

XDrein = Ex8

XDair = ExA10


{--------------------}
}Total exergy destruction rate{
{--------------------}

XDsum = XDp + XDt + XDihe + XDevp + XDcond + XDrein + XDair

XDtot = Ex7 - W_net


{--------------------}
}Performance summary variables{
{--------------------}

etaTh = 100*W_net/Q_ORC
```

```
etaEx = 100*W_net/Ex7

mdot_wf = m_wf

Wnet_A2 = W_net

XDtot_A2 = XDtot


{--------------------}
}Additional useful values{
{--------------------}

etaPlant = 100*W_net/(m_geo*(h[7]-h0))

etaExDrop = 100*W_net/(Ex7-Ex8)


{================================================}
}Part B and C: Heat exchanger areas{
{================================================}

UcondC = 250
UevapC = 850
UiheC = 300


{--------------------}
}Condenser area{
{--------------------}

DTc1 = T[6] - T[10]
DTc2 = T[1] - T[9]

LMTDc = (DTc1 - DTc2)/ln)DTc1/DTc2(

Acnd = Q_cond*1000/(UcondC*LMTDc)


{--------------------}
}Evaporator area{
{--------------------}

DTe1 = T[7] - T[4]
DTe2 = T[8] - T[3]

LMTDe = (DTe1 - DTe2)/ln)DTe1/DTe2(

Aevp = Q_evap*1000/(UevapC*LMTDe)


{--------------------}
}Internal heat exchanger area{
{--------------------}

DTi1 = T[5] - T[3]
DTi2 = T[6] - T[2]

LMTDi = (DTi1 - DTi2)/ln)DTi1/DTi2(

Aihx = Q_IHE*1000/(UiheC*LMTDi)


{================================================}
}Part D-1: Pinch point parametric study outputs{
{================================================}

DTpin_D1 = DT_pinch

Aevap_D1 = Aevp

mwf_D1 = m_wf
```

etaEx_D1 = etaEx

Wnet_D1 = W_net

Qevap_D1 = Q_evap

Tgeo_out_D1 = T[8]

Cpro_D1 = CproGJ

# Appendix F: MATLAB code for secion D_1 plots

```
clc
clear
close all


DT_pinch = [5 7.5 10 12.5 15 17.5 20];


A_evap = [9.707 8.825 8.036 7.324 6.676 6.081 5.532];
m_wf = [1.976 1.887 1.799 1.710 1.621 1.532 1.443];
eta_ex = [41.93 40.05 38.17 36.29 34.40 32.52 30.63];


figure('Color','w','Position',[200 200 850 500])
plot(DT_pinch, A_evap, '-o', 'LineWidth', 1.8, 'MarkerSize', 7)
grid on
box on
xlabel('\DeltaT_{pinch} (^{\circ}C)', 'FontName','Times New Roman',
'FontSize',13)
ylabel('Evaporator area (m^2)', 'FontName','Times New Roman', 'FontSize',13)
title('Effect of Pinch Point Temperature Difference on Evaporator Area', ...
      'FontName','Times New Roman', 'FontSize',14, 'FontWeight','bold')
set(gca,'FontName','Times New Roman','FontSize',12)
exportgraphics(gcf,'D1_evaporator_area_vs_pinch.png','Resolution',300)
figure('Color','w','Position',[200 200 850 500])
plot(DT_pinch, m_wf, '-s', 'LineWidth', 1.8, 'MarkerSize', 7)
grid on
box on
xlabel('\DeltaT_{pinch} (^{\circ}C)', 'FontName','Times New Roman',
'FontSize',13)
ylabel('Working fluid mass flow rate (kg/s)', 'FontName','Times New Roman',
'FontSize',13)
title('Effect of Pinch Point Temperature Difference on Working Fluid Mass Flow
Rate', ...
      'FontName','Times New Roman', 'FontSize',14, 'FontWeight','bold')
set(gca,'FontName','Times New Roman','FontSize',12)
exportgraphics(gcf,'D1_mass_flow_rate_vs_pinch.png','Resolution',300)
figure('Color','w','Position',[200 200 850 500])
plot(DT_pinch, eta_ex, '-^', 'LineWidth', 1.8, 'MarkerSize', 7)
grid on
box on
xlabel('\DeltaT_{pinch} (^{\circ}C)', 'FontName','Times New Roman',
'FontSize',13)
ylabel('Exergy efficiency (%)', 'FontName','Times New Roman', 'FontSize',13)
title('Effect of Pinch Point Temperature Difference on Exergy Efficiency', ...
      'FontName','Times New Roman', 'FontSize',14, 'FontWeight','bold')
set(gca,'FontName','Times New Roman','FontSize',12)
exportgraphics(gcf,'D1_exergy_efficiency_vs_pinch.png','Resolution',300)
result_table = table(DT_pinch', A_evap', m_wf', eta_ex', ...
    'VariableNames', {'DT_pinch_C','A_evap_m2','m_wf_kg_s','eta_ex_percent'});
disp(result_table)
```

# Appendix G: EES code for secion D_2

```
{Input data}
{--------------------}

P0 = 100
T0 = 25

T_cond = 40
T_evp = 120
T_geo = 180

{DT_pinch is defined in the Parametric Table}
{DT_pinch = 10}

eta_T = 0.80
eta_P = 0.90

eps_IHE = 0.60

m_geo = 1


{--------------------}
{Dead state}
{--------------------}

h0 = enthalpy(Water,T=T0,P=P0)
s0 = entropy(Water,T=T0,P=P0)


{--------------------}
{Cycle pressures}
{--------------------}

P[1] = pressure(R123,T=T_cond,x=0)
P[2] = pressure(R123,T=T_evp,x=0)

P[3] = P[2]
P[4] = P[2]

P[5] = P[1]
P[6] = P[1]

P[9] = P0
P[10] = P0


{--------------------}
{State 1: condenser outlet / pump inlet}
{--------------------}

T[1] = T_cond
x[1] = 0

h[1] = enthalpy(R123,T=T[1],x=x[1])
s[1] = entropy(R123,T=T[1],x=x[1])
v[1] = volume(R123,T=T[1],x=x[1])


{--------------------}
{State 2: pump outlet / IHE cold-side inlet}
{--------------------}

h_2s = h[1] + v[1]*(P[2]-P[1])
h[2] = h[1] + (h_2s-h[1])/eta_P

T[2] = temperature(R123,P=P[2],h=h[2])
s[2] = entropy(R123,P=P[2],h=h[2])


{--------------------}
{State 4: evaporator outlet / turbine inlet}
```

```
{--------------------}

T[4] = T_evp
x[4] = 1

h[4] = enthalpy(R123,T=T[4],x=x[4])
s[4] = entropy(R123,T=T[4],x=x[4])


{--------------------}
{State 5: turbine outlet / IHE hot-side inlet}
{--------------------}

s_5s = s[4]
h_5s = enthalpy(R123,P=P[5],s=s_5s)

h[5] = h[4] - eta_T*(h[4]-h_5s)

T[5] = temperature(R123,P=P[5],h=h[5])
s[5] = entropy(R123,P=P[5],h=h[5])


{--------------------}
{State 6: IHE hot-side outlet / condenser inlet}
{--------------------}

T[6] = T[5] - eps_IHE*(T[5]-T[2])

h[6] = enthalpy(R123,P=P[6],T=T[6])
s[6] = entropy(R123,P=P[6],T=T[6])


{--------------------}
{State 3: IHE cold-side outlet / evaporator inlet}
{--------------------}

h[3] = h[2] + h[5] - h[6]

T[3] = temperature(R123,P=P[3],h=h[3])
s[3] = entropy(R123,P=P[3],h=h[3])


{--------------------}
{Geofluid side}
{--------------------}

T[7] = T_geo
x_geo_in = 0

P[7] = pressure(Water,T=T[7],x=x_geo_in)
P[8] = P[7]

h[7] = enthalpy(Water,T=T[7],x=x_geo_in)
s[7] = entropy(Water,T=T[7],x=x_geo_in)

T_pp = T_evp + DT_pinch
h_pp = enthalpy(Water,T=T_pp,P=P[7])

h_f_evp = enthalpy(R123,T=T_evp,x=0)

m_wf*(h[4]-h_f_evp) = m_geo*(h[7]-h_pp)

Q_evap = m_wf*(h[4]-h[3])

h[8] = h[7] - Q_evap/m_geo

T[8] = temperature(Water,P=P[8],h=h[8])
s[8] = entropy(Water,P=P[8],h=h[8])


{--------------------}
{Cooling air side}
{--------------------}
```

```
T[9] = T0
T[10] = T0 + 10

h[9] = enthalpy(Air,T=T[9])
h[10] = enthalpy(Air,T=T[10])

s[9] = entropy(Air,T=T[9],P=P[9])
s[10] = entropy(Air,T=T[10],P=P[10])

Q_cond = m_wf*(h[6]-h[1])

m_air = Q_cond/(h[10]-h[9])


{--------------------}
{Performance parameters}
{--------------------}

W_T = m_wf*(h[4]-h[5])
W_P = m_wf*(h[2]-h[1])
W_net = W_T - W_P

Q_IHE = m_wf*(h[5]-h[6])
Q_ORC = m_wf*(h[4]-h[3])

eta_ORC = W_net/Q_ORC


{--------------------}
{State point output variables}
{--------------------}

T_1 = T[1]
T_2 = T[2]
T_3 = T[3]
T_4 = T[4]
T_5 = T[5]
T_6 = T[6]
T_7 = T[7]
T_8 = T[8]
T_9 = T[9]
T_10 = T[10]

P_1 = P[1]
P_2 = P[2]
P_3 = P[3]
P_4 = P[4]
P_5 = P[5]
P_6 = P[6]
P_7 = P[7]
P_8 = P[8]
P_9 = P[9]
P_10 = P[10]

h_1 = h[1]
h_2 = h[2]
h_3 = h[3]
h_4 = h[4]
h_5 = h[5]
h_6 = h[6]
h_7 = h[7]
h_8 = h[8]
h_9 = h[9]
h_10 = h[10]

s_1 = s[1]
s_2 = s[2]
s_3 = s[3]
s_4 = s[4]
s_5 = s[5]
s_6 = s[6]
s_7 = s[7]
s_8 = s[8]
s_9 = s[9]
```

```
s_10 = s[10]


{--------------------}
{Reference values}
{--------------------}

P_cond = P[1]
P_evp = P[4]

T_dead = T0
P_dead = P0
h_dead = h0
s_dead = s0


T0K = T0 + 273.15


{--------------------}
{Specific flow exergy of geofluid}
{--------------------}

psi7 = (h[7]-h0) - T0K*(s[7]-s0)
psi8 = (h[8]-h0) - T0K*(s[8]-s0)

Ex7 = m_geo*psi7
Ex8 = m_geo*psi8


{--------------------}
{Specific flow exergy of cooling air}
{--------------------}

hAir0 = enthalpy(Air,T=T0)
sAir0 = entropy(Air,T=T0,P=P0)

psiA9 = (h[9]-hAir0) - T0K*(s[9]-sAir0)
psiA10 = (h[10]-hAir0) - T0K*(s[10]-sAir0)

ExA9 = m_air*psiA9
ExA10 = m_air*psiA10


{--------------------}
{Component exergy destruction rates}
{--------------------}

XDp = T0K*m_wf*(s[2]-s[1])

XDt = T0K*m_wf*(s[5]-s[4])

XDihe = T0K*m_wf*((s[3]-s[2]) + (s[6]-s[5]))

XDevp = T0K*(m_wf*(s[4]-s[3]) + m_geo*(s[8]-s[7]))

XDcond = T0K*(m_wf*(s[1]-s[6]) + m_air*(s[10]-s[9]))

XDrein = Ex8

XDair = ExA10


{--------------------}
{Total exergy destruction rate}
{--------------------}

XDsum = XDp + XDt + XDihe + XDevp + XDcond + XDrein + XDair

XDtot = Ex7 - W_net


{--------------------}
{Performance summary variables}
```

```
{--------------------}

etaTh = 100*W_net/Q_ORC

etaEx = 100*W_net/Ex7

mdot_wf = m_wf

Wnet_A2 = W_net

XDtot_A2 = XDtot


{--------------------}
{Additional useful values}
{--------------------}

etaPlant = 100*W_net/(m_geo*(h[7]-h0))

etaExDrop = 100*W_net/(Ex7-Ex8)

UcondC = 250
UevapC = 850
UiheC = 300

{--------------------}
{Condenser area}
{--------------------}

DTc1 = T[6] - T[10]
DTc2 = T[1] - T[9]

LMTDc = (DTc1 - DTc2)/ln(DTc1/DTc2)

Acnd = Q_cond*1000/(UcondC*LMTDc)

{--------------------}
{Evaporator area}
{--------------------}

DTe1 = T[7] - T[4]
DTe2 = T[8] - T[3]

LMTDe = (DTe1 - DTe2)/ln(DTe1/DTe2)

Aevp = Q_evap*1000/(UevapC*LMTDe)

{--------------------}
{Internal heat exchanger area}
{--------------------}

DTi1 = T[5] - T[3]
DTi2 = T[6] - T[2]

LMTDi = (DTi1 - DTi2)/ln(DTi1/DTi2)

Aihx = Q_IHE*1000/(UiheC*LMTDi)


DTpin_D1 = DT_pinch

Aevap_D1 = Aevp

mwf_D1 = m_wf

etaEx_D1 = etaEx

Wnet_D1 = W_net

Qevap_D1 = Q_evap

Tgeo_out_D1 = T[8]

Cpro_D1 = CproGJ
```

# Appendix H: MATLAB code for secion D_2 plots

```
clc
clear
close all


T_evp = [120 125 130 135 140];


eta_th = [13.78 14.31 14.82 15.30 15.75];
eta_ex = [38.17 37.24 35.86 33.98 31.58];


figure('Color','w','Position',[200 200 850 500])
plot(T_evp, eta_th, '-o', 'LineWidth', 1.8, 'MarkerSize', 7)
grid on
box on
xlabel('Evaporator temperature, T_{evp} (^{\circ}C)', ...
       'FontName','Times New Roman', 'FontSize',13)
ylabel('Thermal efficiency (%)', ...
       'FontName','Times New Roman', 'FontSize',13)
title('Effect of Evaporator Temperature on Thermal Efficiency', ...
      'FontName','Times New Roman', 'FontSize',14, 'FontWeight','bold')
set(gca,'FontName','Times New Roman','FontSize',12)
exportgraphics(gcf,'D2_thermal_efficiency_vs_Tevp.png','Resolution',300)


figure('Color','w','Position',[200 200 850 500])
plot(T_evp, eta_ex, '-s', 'LineWidth', 1.8, 'MarkerSize', 7)
grid on
box on
xlabel('Evaporator temperature, T_{evp} (^{\circ}C)', ...
       'FontName','Times New Roman', 'FontSize',13)
ylabel('Exergy efficiency (%)', ...
       'FontName','Times New Roman', 'FontSize',13)
title('Effect of Evaporator Temperature on Exergy Efficiency', ...
      'FontName','Times New Roman', 'FontSize',14, 'FontWeight','bold')
set(gca,'FontName','Times New Roman','FontSize',12)
exportgraphics(gcf,'D2_exergy_efficiency_vs_Tevp.png','Resolution',300)
```

# Appendix I: EES code for secion D_3

```
{--------------------}
{Input data}
{--------------------}

P0 = 100
T0 = 25

{T_cond = 40}
T_evp = 120
T_geo = 180

DT_pinch = 10

eta_T = 0.80
eta_P = 0.90

eps_IHE = 0.60

m_geo = 1


{--------------------}
{Dead state}
{--------------------}

h0 = enthalpy(Water,T=T0,P=P0)
s0 = entropy(Water,T=T0,P=P0)


{--------------------}
{Cycle pressures}
{--------------------}

P[1] = pressure(R123,T=T_cond,x=0)
P[2] = pressure(R123,T=T_evp,x=0)

P[3] = P[2]
P[4] = P[2]

P[5] = P[1]
P[6] = P[1]

P[9] = P0
P[10] = P0


{--------------------}
{State 1: condenser outlet / pump inlet}
{--------------------}

T[1] = T_cond
x[1] = 0

h[1] = enthalpy(R123,T=T[1],x=x[1])
s[1] = entropy(R123,T=T[1],x=x[1])
v[1] = volume(R123,T=T[1],x=x[1])


{--------------------}
{State 2: pump outlet / IHE cold-side inlet}
{--------------------}

h_2s = h[1] + v[1]*(P[2]-P[1])
h[2] = h[1] + (h_2s-h[1])/eta_P

T[2] = temperature(R123,P=P[2],h=h[2])
s[2] = entropy(R123,P=P[2],h=h[2])


{--------------------}
{State 4: evaporator outlet / turbine inlet}
```

```
{--------------------}

T[4] = T_evp
x[4] = 1

h[4] = enthalpy(R123,T=T[4],x=x[4])
s[4] = entropy(R123,T=T[4],x=x[4])


{--------------------}
{State 5: turbine outlet / IHE hot-side inlet}
{--------------------}

s_5s = s[4]
h_5s = enthalpy(R123,P=P[5],s=s_5s)

h[5] = h[4] - eta_T*(h[4]-h_5s)

T[5] = temperature(R123,P=P[5],h=h[5])
s[5] = entropy(R123,P=P[5],h=h[5])


{--------------------}
{State 6: IHE hot-side outlet / condenser inlet}
{--------------------}

T[6] = T[5] - eps_IHE*(T[5]-T[2])

h[6] = enthalpy(R123,P=P[6],T=T[6])
s[6] = entropy(R123,P=P[6],T=T[6])


{--------------------}
{State 3: IHE cold-side outlet / evaporator inlet}
{--------------------}

h[3] = h[2] + h[5] - h[6]

T[3] = temperature(R123,P=P[3],h=h[3])
s[3] = entropy(R123,P=P[3],h=h[3])


{--------------------}
{Geofluid side}
{--------------------}

T[7] = T_geo
x_geo_in = 0

P[7] = pressure(Water,T=T[7],x=x_geo_in)
P[8] = P[7]

h[7] = enthalpy(Water,T=T[7],x=x_geo_in)
s[7] = entropy(Water,T=T[7],x=x_geo_in)

T_pp = T_evp + DT_pinch
h_pp = enthalpy(Water,T=T_pp,P=P[7])

h_f_evp = enthalpy(R123,T=T_evp,x=0)

m_wf*(h[4]-h_f_evp) = m_geo*(h[7]-h_pp)

Q_evap = m_wf*(h[4]-h[3])

h[8] = h[7] - Q_evap/m_geo

T[8] = temperature(Water,P=P[8],h=h[8])
s[8] = entropy(Water,P=P[8],h=h[8])


{--------------------}
{Cooling air side}
{--------------------}
```

```
T[9] = T0
T[10] = T0 + 10

h[9] = enthalpy(Air,T=T[9])
h[10] = enthalpy(Air,T=T[10])

s[9] = entropy(Air,T=T[9],P=P[9])
s[10] = entropy(Air,T=T[10],P=P[10])

Q_cond = m_wf*(h[6]-h[1])

m_air = Q_cond/(h[10]-h[9])


{--------------------}
{Performance parameters}
{--------------------}

W_T = m_wf*(h[4]-h[5])
W_P = m_wf*(h[2]-h[1])
W_net = W_T - W_P

Q_IHE = m_wf*(h[5]-h[6])
Q_ORC = m_wf*(h[4]-h[3])

eta_ORC = W_net/Q_ORC


{--------------------}
{State point output variables}
{--------------------}

T_1 = T[1]
T_2 = T[2]
T_3 = T[3]
T_4 = T[4]
T_5 = T[5]
T_6 = T[6]
T_7 = T[7]
T_8 = T[8]
T_9 = T[9]
T_10 = T[10]

P_1 = P[1]
P_2 = P[2]
P_3 = P[3]
P_4 = P[4]
P_5 = P[5]
P_6 = P[6]
P_7 = P[7]
P_8 = P[8]
P_9 = P[9]
P_10 = P[10]

h_1 = h[1]
h_2 = h[2]
h_3 = h[3]
h_4 = h[4]
h_5 = h[5]
h_6 = h[6]
h_7 = h[7]
h_8 = h[8]
h_9 = h[9]
h_10 = h[10]

s_1 = s[1]
s_2 = s[2]
s_3 = s[3]
s_4 = s[4]
s_5 = s[5]
s_6 = s[6]
s_7 = s[7]
s_8 = s[8]
s_9 = s[9]
```

```
s_10 = s[10]


{--------------------}
{Reference values}
{--------------------}

P_cond = P[1]
P_evp = P[4]

T_dead = T0
P_dead = P0
h_dead = h0
s_dead = s0


{=====================================================}
{Part A-2: Performance summary}
{=====================================================}

T0K = T0 + 273.15


{--------------------}
{Specific flow exergy of geofluid}
{--------------------}

psi7 = (h[7]-h0) - T0K*(s[7]-s0)
psi8 = (h[8]-h0) - T0K*(s[8]-s0)

Ex7 = m_geo*psi7
Ex8 = m_geo*psi8


{--------------------}
{Specific flow exergy of cooling air}
{--------------------}

hAir0 = enthalpy(Air,T=T0)
sAir0 = entropy(Air,T=T0,P=P0)

psiA9 = (h[9]-hAir0) - T0K*(s[9]-sAir0)
psiA10 = (h[10]-hAir0) - T0K*(s[10]-sAir0)

ExA9 = m_air*psiA9
ExA10 = m_air*psiA10


{--------------------}
{Component exergy destruction rates}
{--------------------}

XDp = T0K*m_wf*(s[2]-s[1])

XDt = T0K*m_wf*(s[5]-s[4])

XDihe = T0K*m_wf*((s[3]-s[2]) + (s[6]-s[5]))

XDevp = T0K*(m_wf*(s[4]-s[3]) + m_geo*(s[8]-s[7]))

XDcond = T0K*(m_wf*(s[1]-s[6]) + m_air*(s[10]-s[9]))

XDrein = Ex8

XDair = ExA10


{--------------------}
{Total exergy destruction rate}
{--------------------}

XDsum = XDp + XDt + XDihe + XDevp + XDcond + XDrein + XDair

XDtot = Ex7 - W_net
```

```
{--------------------}
{Performance summary variables}
{--------------------}

etaTh = 100*W_net/Q_ORC

etaEx = 100*W_net/Ex7

mdot_wf = m_wf

Wnet_A2 = W_net

XDtot_A2 = XDtot


{--------------------}
{Additional useful values}
{--------------------}

etaPlant = 100*W_net/(m_geo*(h[7]-h0))

etaExDrop = 100*W_net/(Ex7-Ex8)


{================================================}
{Part B and C: Heat exchanger areas}
{================================================}

UcondC = 250
UevapC = 850
UiheC = 300


{--------------------}
{Condenser area}
{--------------------}

DTc1 = T[6] - T[10]
DTc2 = T[1] - T[9]

LMTDc = (DTc1 - DTc2)/ln(DTc1/DTc2)

Acnd = Q_cond*1000/(UcondC*LMTDc)


{--------------------}
{Evaporator area}
{--------------------}

DTe1 = T[7] - T[4]
DTe2 = T[8] - T[3]

LMTDe = (DTe1 - DTe2)/ln(DTe1/DTe2)

Aevp = Q_evap*1000/(UevapC*LMTDe)


{--------------------}
{Internal heat exchanger area}
{--------------------}

DTi1 = T[5] - T[3]
DTi2 = T[6] - T[2]

LMTDi = (DTi1 - DTi2)/ln(DTi1/DTi2)

Aihx = Q_IHE*1000/(UiheC*LMTDi)


oph = 7000
cgf = 1.3
life = 20
```

```
intr = 0.14
mf = 1.06

CRF = intr*(1 + intr)^life/((1 + intr)^life - 1)


{--------------------}
{Geofluid exergy input}
{--------------------}

T0KE = T0 + 273.15

psiGF = (h[7] - h0) - T0KE*(s[7] - s0)

ExGF = m_geo*psiGF


{--------------------}
{Capital investment cost of components}
{--------------------}

Ztur = 4405*W_T^0.89

Zpmp = 1120*W_P^0.8

Zcnd = 1397*Acnd^0.89

Zevp = 1397*Aevp^0.89

Zihx = 2143*Aihx^0.514

Ztot = Ztur + Zpmp + Zcnd + Zevp + Zihx


{--------------------}
{Cost rates of components}
{--------------------}

ZdotTur = Ztur*mf*CRF/(oph*3600)

ZdotPmp = Zpmp*mf*CRF/(oph*3600)

ZdotCnd = Zcnd*mf*CRF/(oph*3600)

ZdotEvp = Zevp*mf*CRF/(oph*3600)

ZdotIhx = Zihx*mf*CRF/(oph*3600)

ZdotTot = ZdotTur + ZdotPmp + ZdotCnd + ZdotEvp + ZdotIhx


{--------------------}
{Fuel cost rate}
{--------------------}

Cfuel = cgf*ExGF/1000000


{--------------------}
{Total product cost rate and specific production cost}
{--------------------}

CdotPro = Cfuel + ZdotTot

CproGJ = CdotPro/W_net*1000000

CproKWh = CproGJ*0.0036

CdotPro_h = CdotPro*3600


{================================================}
{Part D-3: Condenser temperature parametric study outputs}
```

{================================================}

Tcond_D3 = T_cond

Pcond_D3 = P_cond

etaTh_D3 = etaTh

etaEx_D3 = etaEx

CdotPro_D3 = CdotPro

CdotProh_D3 = CdotPro_h

CproGJ_D3 = CproGJ

CproKWh_D3 = CproKWh

Wnet_D3 = W_net

Qcond_D3 = Q_cond

Acnd_D3 = Acnd

mwf_D3 = m_wf

# Appendix J: MATLAB code for secion D_3 plots

```
clc
clear
close all


T_cond = [30 35 40 45 50];


eta_th = [15.35 14.57 13.78 12.98 12.19];
eta_ex = [44.57 41.32 38.17 35.11 32.15];


CdotPro = [0.002493 0.001912 0.001651 0.001482 0.001353];
CdotPro_h = CdotPro * 3600;


figure('Color','w','Position',[200 200 850 500])
plot(T_cond, eta_th, '-o', 'LineWidth', 1.8, 'MarkerSize', 7)
grid on
box on
xlabel('Condenser temperature, T_{cond} (^{\circ}C)', ...
       'FontName','Times New Roman', 'FontSize',13)
ylabel('Thermal efficiency (%)', ...
       'FontName','Times New Roman', 'FontSize',13)
title('Effect of Condenser Temperature on Thermal Efficiency', ...
      'FontName','Times New Roman', 'FontSize',14, 'FontWeight','bold')
set(gca,'FontName','Times New Roman','FontSize',12)
exportgraphics(gcf,'D3_thermal_efficiency_vs_Tcond.png','Resolution',300)




figure('Color','w','Position',[200 200 850 500])
plot(T_cond, eta_ex, '-s', 'LineWidth', 1.8, 'MarkerSize', 7)
grid on
box on
xlabel('Condenser temperature, T_{cond} (^{\circ}C)', ...
       'FontName','Times New Roman', 'FontSize',13)
ylabel('Exergy efficiency (%)', ...
       'FontName','Times New Roman', 'FontSize',13)
title('Effect of Condenser Temperature on Exergy Efficiency', ...
      'FontName','Times New Roman', 'FontSize',14, 'FontWeight','bold')
set(gca,'FontName','Times New Roman','FontSize',12)
exportgraphics(gcf,'D3_exergy_efficiency_vs_Tcond.png','Resolution',300)




figure('Color','w','Position',[200 200 850 500])
plot(T_cond, CdotPro_h, '-^', 'LineWidth', 1.8, 'MarkerSize', 7)
grid on
```

```
box on
xlabel('Condenser temperature, T_{cond} (^{\circ}C)', ...
       'FontName','Times New Roman', 'FontSize',13)
ylabel('Total product cost rate ($/h)', ...
       'FontName','Times New Roman', 'FontSize',13)
title('Effect of Condenser Temperature on Total Product Cost Rate', ...
      'FontName','Times New Roman', 'FontSize',14, 'FontWeight','bold')
set(gca,'FontName','Times New Roman','FontSize',12)
exportgraphics(gcf,'D3_product_cost_rate_vs_Tcond.png','Resolution',300)



result_table = table(T_cond', eta_th', eta_ex', CdotPro', CdotPro_h', ...
    'VariableNames', {'T_cond_C','eta_th_percent','eta_ex_percent', ...
                      'CdotPro_dollar_per_s','CdotPro_dollar_per_h'});


disp(result_table)
```